17. Find the first three multipole moments for a single point charge q sitting at the point $\mathbf{r} = R(\mathbf{i} + \mathbf{j} - \mathbf{k})$. Determine, through three terms in the multipole expansion, the potential (in spherical coordinates) at a point very far away from the charge.

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CHAPTER 2

CALCULUS OF VARIATIONS

INTRODUCTION

One of the oldest problems in mathematical physics is that of trying to minimize expressions which do not depend on some simple continuous variable, but rather on a function. That is, the problem of interest is not just to determine the point at which some particular function takes on its smallest value, but to determine the entire functional dependence of some function in such a way that it will minimize an integral involving that function and its derivatives. Such problems played a crucial role in the development of classical physics during the eighteenth and nineteenth centuries, and their importance did not diminish with the arrival of quantum concepts in the twentieth century. This subject is called the calculus of variations, and in this chapter we will discuss briefly the ideas and applications of the calculus of variations to classical physics in order to familiarize the reader with the concepts and techniques involved. In later chapters we will have occasion to return to these ideas in a more modern context.

2.1 SOME FAMOUS PROBLEMS

The calculus of variations is an old subject. The study of calculus itself had been in existence for only a few years before questions were being asked that we now view as part of the subject of the calculus of variations. Three such questions are:

1) What plane curve connecting two given points has the shortest length?

We can formulate this question mathematically as follows. Let (x_A, y_A) and (x_B, y_B) be two points in the xy-plane with $x_A < x_B$. Let $y = y(x)$ connect them—thus, $y(x_A) = y_A$ and $y(x_B) = y_B$. The length of the arc connecting them will be

$$I = \int_{(x_A,\, y_A)}^{(x_B,\, y_B)} ds \, .$$

Since $ds^2 = dx^2 + dy^2$, we have

$$ds = \sqrt{1 + y'^2}\, dx \, ,$$

where y' denotes the derivative of y with respect to x. The problem then is to choose the function $y(x)$, which is buried in the *integrand*, in such a way as to make the *integral* I a minimum.

2) In 1696 Johann Bernoulli of the University of Basel posed the following problem: Given two points A and B in a vertical plane, find the path AMB which the movable particle M will traverse in the shortest time, assuming that its acceleration is due only to gravity. This is the famous *brachistochrone* problem (from the Greek, *brachistos* = shortest, *chronos* = time), and it marks the beginning of general interest in the calculus of variations.

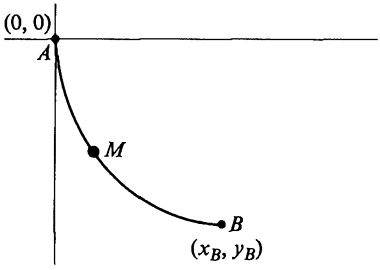

Fig. 2.1 Bernoulli's brachistochrone.

Let us take A as the origin of our coordinate system, assume that the particle of mass m initially has zero velocity (at A) and, as did Bernoulli, assume that there is no friction. We take the y-axis directed vertically downward for convenience. The speed along the curve AMB of Fig. 2.1 is $v = ds/dt$, so the total time of descent is

$$I = \int_A^B \frac{ds}{v} = \int_{x_A=0}^{x_B} \frac{\sqrt{1+y'^2}}{v}\, dx\,.$$

To complete the specification of the problem, we must compute v as a function of position. We know that the increase of kinetic energy is equal to the decrease of potential energy if there is no friction. Hence

$$\tfrac{1}{2}mv^2 = mgy$$

so that

$$I = \frac{1}{\sqrt{2g}} \int_0^{x_B} \sqrt{\frac{1+y'^2}{y}}\, dx\,.$$

The problem again is to find the function $y(x)$ which minimizes I.

3) A third problem is that of finding the minimum surface of revolution passing through two given fixed points, (x_A, y_A) and (x_B, y_B). Let us suppose that the curve $y(x)$ is to be rotated about the x-axis as illustrated in Fig. 2.2. We want to minimize the area of the surface so generated. We assume that $y_A > 0$, $y_B > 0$ and $y(x) \geq 0$ for $x_A \leq x \leq x_B$ as shown in the diagram. The integral we seek to minimize is the surface area:

$$I = 2\pi \int_A^B y\, ds = 2\pi \int_{x_A}^{x_B} y\sqrt{1+y'^2}\, dx\,,$$

where $y(x_A) = y_A$ and $y(x_B) = y_B$.

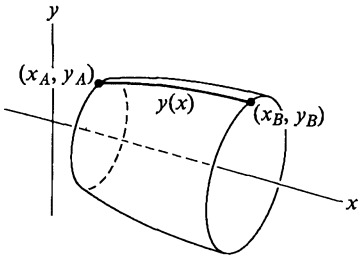

Fig. 2.2 The minimum surface of revolution.

All three of these problems are special cases of the more general question: What is the function $y(x)$, with $y(x_A) = y_A$ and $y(x_B) = y_B$ which minimizes the integral

$$I = \int_{x_A}^{x_B} f(x, y, y')\, dx\,,$$

where f is a *given* function of x, y, and y'? In geometrical language we are looking for the *path of integration* from x_A to x_B which minimizes the integral I.

2.2 THE EULER-LAGRANGE EQUATION

In this section we derive the differential equation that $y(x)$ must obey in order to minimize the integral

$$I = \int_{x_A}^{x_B} f(x, y, y')\, dx\,.$$

x_A, x_B, $y(x_A) = y_A$, $y(x_B) = y_B$ and f are all given, and f is assumed to be a twice-differentiable function of all its arguments. We denote the function which minimizes I by $y(x)$ (see Fig. 2.3) and consider the one-parameter family of comparison functions (or test functions), $\bar{y}(x, \epsilon)$, which satisfy the conditions:

a) $\bar{y}(x_A, \epsilon) = y_A$, $\bar{y}(x_B, \epsilon) = y_B$ for all ϵ;

b) $\bar{y}(x, 0) = y(x)$, the desired minimizing function;

c) $\bar{y}(x, \epsilon)$ and all its derivatives through second order are continuous functions of x and ϵ.

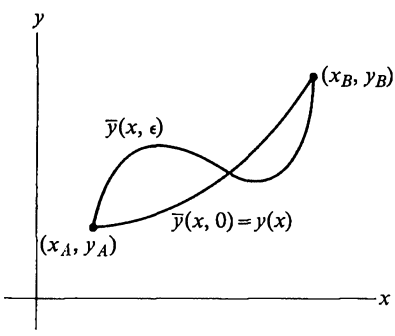

Fig. 2.3 A one-parameter family of comparison functions.

For a given comparison function, the integral

$$I(\epsilon) = \int_{x_A}^{x_B} f(x, \bar{y}, \bar{y}')\, dx$$

is clearly a function of ϵ. Also, since setting $\epsilon = 0$ corresponds, by condition (b), to replacing $\bar{y}$ by $y(x)$ and $\bar{y}'$ by $y'(x)$, we see that $I(\epsilon)$ should be a minimum with respect to ϵ for the value $\epsilon = 0$ according to the designation that $y(x)$ is the actual minimizing function. This is true for *any* $\bar{y}(x, \epsilon)$.

What we are doing here is considering a family of "smooth" curves passing through the points (x_A, y_A) and (x_B, y_B) which are labeled by the variable ϵ. The curve which makes the number I as small as possible has the label $\epsilon = 0$. We should emphasize that the choice of $\epsilon = 0$ is entirely arbitrary. We could equally well designate the minimizing function by the label $\epsilon = 1.67$ and change condition (b) accordingly. Since this curve by assumption minimizes I, and since I depends continuously on ϵ and is a differentiable function of ϵ [this follows from condition (c)], we can apply the standard methods of elementary calculus to determine the minimum.

A necessary, *but not sufficient*, condition for a minimum is the vanishing of the first derivative. Thus we have

$$\left[\frac{dI}{d\epsilon}\right]_{\epsilon=0} = 0 \tag{2.1}$$

as a necessary condition for the integral to take on a minimum value at $\epsilon = 0$. We now differentiate $I(\epsilon)$ with respect to ϵ (remember, x is not a function of ϵ, only $\bar{y}$ and $\bar{y}'$ are):

$$\frac{dI}{d\epsilon} = \int_{x_A}^{x_B} \left[\frac{\partial f}{\partial \bar{y}} \frac{d\bar{y}}{d\epsilon} + \frac{\partial f}{\partial \bar{y}'} \frac{d\bar{y}'}{d\epsilon}\right] dx$$

which we may write according to condition (c) as

$$\frac{dI}{d\epsilon} = \int_{x_A}^{x_B} \left[\frac{\partial f}{\partial \bar{y}} \frac{d\bar{y}}{d\epsilon} + \frac{\partial f}{\partial \bar{y}'} \frac{d}{dx}\left(\frac{d\bar{y}}{d\epsilon}\right)\right] dx\,.$$

Now, if we integrate the second term by parts, we will obtain

$$\frac{dI}{d\epsilon} = \int_{x_A}^{x_B} \frac{\partial f}{\partial \bar{y}} \frac{d\bar{y}}{d\epsilon}\, dx + \left[\frac{d\bar{y}}{d\epsilon} \frac{\partial f}{\partial \bar{y}'}\right]_{x_A}^{x_B} - \int_{x_A}^{x_B} \frac{d\bar{y}}{d\epsilon} \frac{d}{dx}\left(\frac{\partial f}{\partial \bar{y}'}\right) dx\,.$$

But by condition (a), $\bar{y}(x_A, \epsilon) = y_A$ and $\bar{y}(x_B, \epsilon) = y_B$ for *all* ϵ. Hence

$$\left.\frac{d\bar{y}}{d\epsilon}\right|_{x=x_A} = 0 = \left.\frac{d\bar{y}}{d\epsilon}\right|_{x=x_B}$$

so we get, finally,

$$\frac{dI}{d\epsilon} = \int_{x_A}^{x_B} \left[\frac{\partial f}{\partial \bar{y}} - \frac{d}{dx}\left(\frac{\partial f}{\partial \bar{y}'}\right)\right] \frac{d\bar{y}}{d\epsilon}\, dx\,.$$

We now require that $I(\epsilon)$ have a minimum at $\epsilon = 0$, that is,

$$\left[\frac{dI}{d\epsilon}\right]_{\epsilon=0} = 0 = \int_{x_A}^{x_B}\left[\frac{\partial f}{\partial \bar{y}} - \frac{d}{dx}\left(\frac{\partial f}{\partial \bar{y}'}\right)\right]_{\epsilon=0}\left[\frac{d\bar{y}}{d\epsilon}\right]_{\epsilon=0} dx .$$

Now setting $\epsilon = 0$ is equivalent to setting $\bar{y}(x, \epsilon) = y(x)$, $\bar{y}'(x, \epsilon) = y'(x)$, and $\bar{y}''(x, \epsilon) = y''(x)$. [Note that the integrand does depend on $\bar{y}''$, and in taking the limit $\epsilon = 0$, we need to know that the second derivative $y''(x, \epsilon)$ is a continuous function of its two variables. This is guaranteed by condition (c).] Calling

$$\left[\frac{d\bar{y}}{d\epsilon}\right]_{\epsilon=0} = \eta(x) ,$$

we obtain

$$\int_{x_A}^{x_B}\left[\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)\right]\eta(x)\, dx = 0 . \tag{2.2}$$

But $\eta(x)$, apart from the fact that it vanishes at x_A and x_B by condition (a) and is continuous and differentiable by condition (c), is completely arbitrary. Thus for the integral in Eq. (2.2) to vanish, we must have

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0 . \tag{2.3}$$

This equation was derived by Euler in 1744. It is known as the Euler-Lagrange equation because it is also the basis for Lagrange's formulation of classical mechanics. If we expand the total derivative with respect to x, Eq. (2.3) becomes

$$\frac{\partial f}{\partial y} - \frac{\partial^2 f}{\partial x\, \partial y'} - \frac{\partial^2 f}{\partial y\, \partial y'}\, y' - \frac{\partial^2 f}{\partial^2 y'} y'' = 0 ,$$

a second-order differential equation, whose solution supplies the twice-differentiable minimizing function $y(x)$, provided the minimum exists. The condition of Eq. (2.1) is only a *necessary* condition for a minimum; the solution $y(x)$ could also produce a maximum or even a point of inflection of the function $I(\epsilon)$ at $\epsilon = 0$. Thus what we have really found is an *extremizing* function $y(x)$. From the nature of the problem, we can usually tell the nature of the extremum without investigating the sign of higher derivatives of $I(\epsilon)$, which is in general a very complicated procedure. For example, in the three problems discussed in Section 2.1, it is clear on physical grounds that maxima do not exist. Also, there is the question whether we have found a relative or absolute minimum. In general, we shall not investigate these problems in detail mathematically, but shall rely on a physical interpretation of the situation to tell us the type of extremum involved.

In many cases of physical interest, the Euler-Lagrange equation can be greatly simplified. In particular, note that in all the examples considered in

Section 2.1, the function $f(x, y, y')$ does not depend *explicitly* on x, only implicitly through the x-dependence of y and y'. With this in mind, consider the quantity

$$\frac{d}{dx}\left[y'\frac{\partial f}{\partial y'} - f\right].$$

Performing the differentiations, we get

$$\begin{aligned}\frac{d}{dx}\left[y'\frac{\partial f}{\partial y'} - f\right] &= y''\frac{\partial f}{\partial y'} + y'\frac{d}{dx}\frac{\partial f}{\partial y'} - \frac{\partial f}{\partial x} - \frac{\partial f}{\partial y}y' - \frac{\partial f}{\partial y'}y'' \\ &= -y'\left[\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)\right] - \frac{\partial f}{\partial x}.\end{aligned}$$

But if y satisfies the Euler-Lagrange equation, that is, Eq. 2.3, and if f does not depend explicitly on x so that

$$\frac{\partial f}{\partial x} = 0,$$

then we conclude that

$$\frac{d}{dx}\left[y'\frac{\partial f}{\partial y'} - f\right] = 0.$$

We can integrate this equation to get

$$y'\frac{\partial f}{\partial y'} - f = \text{const.} \tag{2.4}$$

Thus we are left with a simple *first-order* differential equation for $y(x)$.

The techniques used in deriving the one-variable Euler-Lagrange equation can readily be extended to integrals of the form

$$I = \int_{x_A}^{x_B} f(x, y_1, y_2, \cdots, y_n; y_1', y_2', \cdots, y_n')\, dx.$$

We want to determine the functions $y_1(x), y_2(x), \cdots, y_n(x)$ which extremize I, subject to the conditions

$$\begin{aligned} y_1(x_A) &= y_{1A}, & y_1(x_B) &= y_{1B} \\ y_2(x_A) &= y_{2A}, & y_2(x_B) &= y_{2B} \\ &\vdots & &\vdots \\ y_n(x_A) &= y_{nA}, & y_n(x_B) &= y_{nB}. \end{aligned}$$

This problem is virtually identical to the one we have just solved. In fact, if we imagine that $(n - 1)$ of the functions $\bar{y}_j$ are given their extremizing form, $y_j(x)$, and if we then vary the one remaining function, say $\bar{y}_i$, in such a way as to extremize I, then the problem does indeed coincide with the extremal problem which we have just solved. The equation satisfied by $y_i(x)$, the ith extremizing

function, is then according to Eq. 2.3, just

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial y_i'}\right) = 0\,, \qquad i = 1, 2, \cdots, n\,. \tag{2.5}$$

Again, just as in the case of one dependent variable, if f does not depend explicitly on x, then we get the first integral

$$\sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'} - f = \text{const} \tag{2.6}$$

which follows immediately by evaluating

$$\frac{d}{dx}\left[\sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'} - f\right]$$

and assuming that the $y_i(x)$ satisfy Eq. (2.5).

Note that as a problem-solving device Eq. (2.6), for the case of several dependent variables, is not nearly as useful as Eq. (2.4), for the case of one dependent variable. Equation (2.4) reduces the entire problem to solving a single first-order differential equation while Eq. (2.6) provides only an equation linking the first derivatives of *all* the y_i. In this case one still must come to grips with some second-order differential equations.

2.3 SOME FAMOUS SOLUTIONS

In this section we shall discuss the solutions to the three classic problems posed in Section 2.1 in the order in which they were presented there.

Problem 1. The shortest line. In this case, $f = \sqrt{1 + y'^2}$. Applying the Euler-Lagrange equation directly, we get, since $\partial f/\partial y = 0$,

$$\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0 \qquad \text{or} \qquad \frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}} = \text{const.}$$

But this just tells us that y' equals some constant, say $y' = A$, so that

$$y = Ax + B\,,$$

where A and B are constants which can be chosen so that the curve, a straight line, passes through the fixed endpoints.

In this case f does not depend explicitly on x so we can equally well use Eq. (2.4),

$$y' \frac{\partial f}{\partial y'} - f = \text{const.}$$

Using the given form of f, we get immediately

$$\frac{1}{\sqrt{1 + y'^2}} = \text{const,}$$

from which we conclude that y' is a constant, so again we get $y = Ax + B$.

Problem 2. The brachistochrone. For this problem, we saw in Section 2.1 that

$$f = \sqrt{\frac{1 + y'^2}{y}}$$

(we have dropped an irrelevant factor of $1/\sqrt{2g}$). Since f does not depend explicitly on x, we use Eq. (2.4) to get

$$\frac{y'^2}{\sqrt{y(1 + y'^2)}} - \sqrt{\frac{1 + y'^2}{y}} = C \qquad \text{or} \qquad \frac{1}{y(1 + y'^2)} = C^2 .$$

Letting $1/C^2 = 2a$, we get

$$y' = \sqrt{\frac{2a - y}{y}} ,$$

and integration yields

$$x - x_0 = \int \sqrt{\frac{y}{2a - y}}\, dy .$$

If we make the change of variable $y = a(1 - \cos\theta)$, then the integral becomes

$$x - x_0 = 2a \int \sin^2 \frac{\theta}{2}\, d\theta = a(\theta - \sin\theta) .$$

Thus the solution to the brachistochrone problem is, in parametric form,

$$x = a(\theta - \sin\theta) + x_0 \qquad y = a(1 - \cos\theta) .$$

These are the equations of a cycloid generated by the motion of a fixed point on the circumference of a circle of radius a which rolls on the positive side of the line $y = 0$, that is, on the underside of the x-axis of Fig. 2.1. There exists one and only one cycloid through the origin and the point (x_B, y_B); a suitable choice of a and x_0 will give this cycloid.

When Bernoulli first solved the brachistochrone problem, he did not do it in this way. Although he boasted of having discovered a wonderful solution, he did not publish it immediately, but instead challenged other mathematicians —especially his elder brother, Jacob, with whom he had been carrying on a bitter feud, and had publicly characterized as an incompetent—to find the answer. He finally published his result in 1697.* Perhaps it was Bernoulli who first said, "It is not enough for you to succeed; your colleagues must fail."

The cycloid had been discovered only a short time before Bernoulli's work in connection with mechanical problems involving the ideal pendulum. (An ideal pendulum is one whose period is exactly independent of the amplitude of oscillation. For a pendulum whose bob describes a circular path—the only one that is easy to design—the period is strictly independent of amplitude only in the limit of zero amplitude.) Huygens had discovered that a mass point

* See Courant and Robbins, *What is Mathematics?*, pp. 379–384.

oscillating without friction under the influence of gravity on a vertical cycloid has a period independent of amplitude; thus the cycloid had been called the tautochrone. With Bernoulli's discovery, this curve was rechristened the brachistochrone.

Problem 3. The minimum surface. In this case, we found in Section 2.1 that $f = y\sqrt{1 + y'^2}$. Again f does not depend explicitly on x, so

$$y' \frac{\partial f}{\partial y'} - f = -b ,$$

where we write the constant as $-b$ for convenience. Using our particular f, we get immediately

$$\frac{y}{\sqrt{1 + y'^2}} = b$$

from which we obtain

$$x - x_0 = \int \frac{dy}{\sqrt{\frac{y^2}{b^2} - 1}} = b \cosh^{-1}\left(\frac{y}{b}\right) .$$

Thus

$$y = b \cosh \frac{x - x_0}{b} .$$

Since the curve is supposed to lie entirely above the x-axis, we see that we must have $b > 0$.

The curve is a catenary (from the Latin, *catena* = chain); the surface it generates is a catenoid of revolution. The constants b and x_0 must be picked so that the catenary passes through the points (x_A, y_A) and $x(_B, y_B)$ without dropping below the x-axis. This cannot always be done! The theory developed here is deficient, primarily in that it will only determine for us the *twice-differentiable* minimizing curves. Let us examine the problem of connecting the points (x_A, y_A) and (x_B, y_B) in more detail. If the catenary is to pass through (x_A, y_A), then b and x_0 are related by

$$y_A = b \cosh \frac{x_A - x_0}{b} .$$

This tells us that

$$x_0 = x_A - b \cosh^{-1} \frac{y_A}{b} ,$$

so the equation of the catenary becomes

$$y = b \cosh \left[\frac{x - x_A}{b} + \cosh^{-1}\left(\frac{y_A}{b}\right)\right] .$$

This equation describes a one-parameter family of catenaries which pass through the point (x_A, y_A), as shown in Fig. 2.4. In the last equation $\cosh^{-1} y_A/b$ must be chosen negative if the catenaries are to take on minima to the right of x_A.

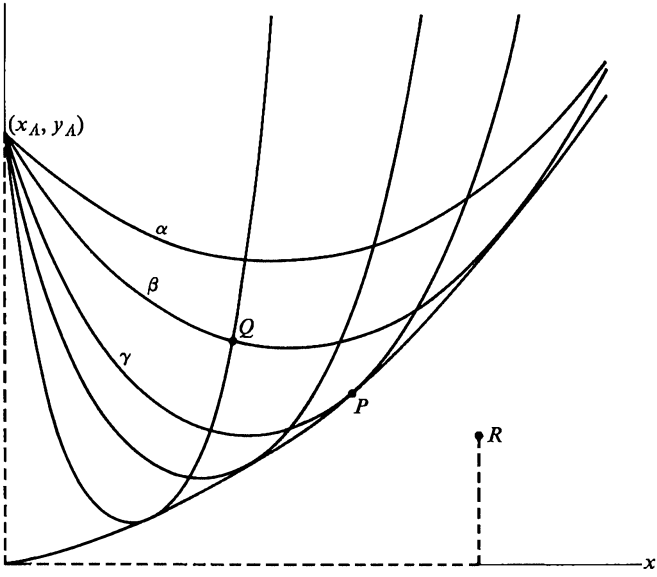

Fig. 2.4

One can show that every catenary $(\alpha, \beta, \gamma, \cdots)$ through (x_A, y_A) which does not intersect the x-axis is tangent to the envelope curve in Fig. 2.4. The envelope curve is a parabola with its focus at (x_A, y_A). No such catenary can ever pass through a point such as R, which is separated from (x_A, y_A) by the envelope. Also, only one catenary passes through a point like P lying on the envelope, but two catenaries pass through a point like Q lying on the same side of the envelope as (x_A, y_A). Thus we see that for a point like R, there is no twice-differentiable miminizing curve. It can be shown that the minimum area is generated in this case by the broken line whose segments are

$$\begin{aligned} x &= x_A && \text{for} \quad 0 \leq y \leq y_A\,, \\ y &= 0 && \text{for} \quad x_A \leq x \leq x_R\,, \\ x &= x_R && \text{for} \quad 0 \leq y \leq y_R\,. \end{aligned}$$

This solution is called the Goldschmidt discontinuous solution. It generates a surface of area equal to $\pi(y_A^2 + y_R^2)$, a spool-shaped surface. This surface is always a relative minimum, but it turns out that there exists a range of points (x_B, y_B) for which the catenary provides the absolute minimum. We enter the range of such points if we go sufficiently far above (or to the left of) the envelope. Then the upper catenary provides an absolute minimum surface of revolution. The upper catenary through a point such as Q always provides at

least a relative minimum. The lower catenary through such a point as Q does not even generate a relative minimum, nor does the (unique) catenary through a point such as P.

We have analyzed in detail this defective "solution" to point out the limitations and the subtleties of the theory if studied rigorously.

2.4 ISOPERIMETRIC PROBLEMS—CONSTRAINTS

It was once the practice to reward exceptional military or civil service in the following way. The king would give to the one being honored all the land that he could encompass by a ploughed furrow in a specified period of time. The problem thus arose of finding the plane curve of given length which encloses the greatest possible area—the isoperimetric problem ("same perimeter").

In mathematical terms, we are trying to derive the differential equation that must be satisfied by the function $y(x)$ which makes the integral

$$I = \int_{x_A}^{x_B} f(x, y, y')\, dx$$

an extremum and also satisfies a second condition, such as giving the integral

$$J = \int_{x_A}^{x_B} g(x, y, y')\, dx$$

a given value with $y(x_A) = y_A$ and $y(x_B) = y_B$ both satisfied. Of course, one could easily imagine a more general class of problems. For example, we could allow for the possibility of more than one dependent variable, say $y_1(x)$, $y_2(x)$, $\cdots$, $y_n(x)$, and we might envision several integrals which have assigned values, $J_1, J_2, \cdots, J_m$. However, for the purposes of a proof, it suffices to illustrate the method on the simple case stated above. We shall then write the more general result, the proof of which we will leave to the reader as an exercise in developing a notation suitable to the increased generality.

The procedure is essentially the same as in Section 2.2. We consider a *two*-parameter family of functions $\bar{y}(x, \epsilon_1, \epsilon_2)$ with the following conditions:

(α) $\bar{y}(x_A, \epsilon_1, \epsilon_2) = y_A$, $\bar{y}(x_B, \epsilon_1, \epsilon_2) = y_B$ for all ϵ_1 and ϵ_2;

(β) $\bar{y}(x, 0, 0) = y(x)$, the desired extremizing function;

(γ) $\bar{y}(x, \epsilon_1, \epsilon_2)$ has continuous derivatives through second order in all variables.

We form

$$I(\epsilon_1, \epsilon_2) = \int_{x_A}^{x_B} f(x, \bar{y}, \bar{y}')\, dx\,,$$

$$J(\epsilon_1, \epsilon_2) = \int_{x_A}^{x_B} g(x, \bar{y}, \bar{y}')\, dx = \text{const.}$$

Just as in Section 2.2, we want to extremize $I(\epsilon_1, \epsilon_2)$ with respect to ϵ_1 and ϵ_2, requiring the extremum to be at $\epsilon_1 = 0$, $\epsilon_2 = 0$, but now subject to the condition that $J(\epsilon_1, \epsilon_2)$ have a fixed value. Clearly, ϵ_1 and ϵ_2 are not independent if

$J(\epsilon_1, \epsilon_2)$ has a fixed value, and it is for this reason that we elected to work with a two-parameter family of functions, $\bar{y}(\epsilon_1, \epsilon_2)$. If we had just $J(\epsilon)$, then the constant value of J would determine ϵ, which would, in turn, determine $I(\epsilon)$.

Now we use the techniques of elementary calculus to form the quantity

$$K(\epsilon_1, \epsilon_2) = I(\epsilon_1, \epsilon_2) + \lambda J(\epsilon_1, \epsilon_2),$$

where λ is a constant—as yet undetermined—referred to as a *Lagrange multiplier*. The condition for an extremum of $K(\epsilon_1, \epsilon_2)$ [and hence of $I(\epsilon_1, \epsilon_2)$] to occur at $\epsilon_1 = 0$, $\epsilon_2 = 0$ is

$$\left[\frac{\partial K}{\partial \epsilon_1}\right]_{\substack{\epsilon_1=0\\ \epsilon_2=0}} = 0 = \left[\frac{\partial K}{\partial \epsilon_2}\right]_{\substack{\epsilon_1=0\\ \epsilon_2=0}}$$

where K is given simply by

$$K(\epsilon_1, \epsilon_2) = \int_{x_A}^{x_B} h(x, \bar{y}, \bar{y}')\, dx\,,$$

and $h = f + \lambda g$. We now must calculate the two first partial derivatives of K:

$$\frac{\partial K}{\partial \epsilon_j} = \int_{x_A}^{x_B}\left[\frac{\partial h}{\partial \bar{y}}\frac{\partial \bar{y}}{\partial \epsilon_j} + \frac{\partial h}{\partial \bar{y}'}\frac{\partial \bar{y}'}{\partial \epsilon_j}\right] dx\,, \qquad j = 1, 2\,.$$

Integrating by parts exactly as in Section 2.2, we get

$$\frac{\partial K}{\partial \epsilon_j} = \int_{x_A}^{x_B}\left[\frac{\partial h}{\partial \bar{y}} - \frac{d}{dx}\left(\frac{\partial h}{\partial \bar{y}'}\right)\right]\frac{\partial \bar{y}}{\partial \epsilon_j}\, dx\,, \qquad j = 1, 2\,,$$

where we have used condition (α) to drop the boundary term. Now we set $\epsilon_1 = 0$, $\epsilon_2 = 0$ and require that the two first partial derivatives of K vanish. Thus

$$\int_{x_A}^{x_B}\left[\frac{\partial h}{\partial y} - \frac{d}{dx}\left(\frac{\partial h}{\partial y'}\right)\right]\eta_j(x)\, dx = 0\,, \qquad j = 1, 2\,,$$

where we have written

$$\left[\frac{\partial \bar{y}}{\partial \epsilon_j}\right]_{\substack{\epsilon_1=0\\ \epsilon_2=0}} = \eta_j(x)\,.$$

But the functions $\eta_1(x)$ and $\eta_2(x)$ are arbitrary, so we conclude that

$$\frac{\partial h}{\partial y} - \frac{d}{dx}\left(\frac{\partial h}{\partial y'}\right) = 0$$

is the equation for the extremizing function $y(x)$.

This looks just like the Euler-Lagrange equation of Section 2.2, except that the quantity $h = f + \lambda g$ replaces f. Note that the solution of the Euler-Lagrange equation involves two constants of integration *plus* the multiplier λ. These three numbers are just enough to ensure that $y(x)$ passes through the two endpoints, (x_A, y_A) and (x_B, y_B), *and* that the integral J has the prescribed value.

The generalizations of this result are not difficult. First, if we have m constraint integrals, $J_1, J_2, \cdots, J_m$, then the function h which obeys the Euler-Lagrange equation is just

$$h = f + \sum_{i=1}^{m} \lambda_i g_i ,$$

where

$$J_i = \int_{x_A}^{x_B} g_i(x, y, y')\, dx = \text{const.}$$

If we want to allow for the possibility of n dependent variables, a simple extension of notation shows that (as indicated in Section 2.2) we obtain the set of Euler-Lagrange equations

$$\frac{\partial h}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial h}{\partial y_i'}\right) = 0 , \qquad i = 1, 2, \cdots, n , \tag{2.7}$$

where again

$$h = f + \sum_{i=1}^{m} \lambda_i g_i . \tag{2.8}$$

The reader can easily go through the few steps necessary to obtain Eqs. (2.7) and (2.8). Note that if there are m constraint integrals, it will be necessary to use an $(m + 1)$-parameter collection of functions, $\bar{y}(x, \epsilon_1, \epsilon_2, \cdots, \epsilon_{m+1})$, in the general case.

We now set about solving the original isoperimetric problem. To find the curve of fixed length L and maximum area, we want to find the functions $x(t)$ and $y(t)$ which maximize

$$A = \tfrac{1}{2}\int_{t_A}^{t_B} (xy' - x'y)\, dt .$$

This is the formula for the area inside a closed curve in the xy-plane, when we parametrize the coordinates x and y by the variable t. The reader should show that in polar coordinates the expression above is equivalent to

$$A = \tfrac{1}{2}\int_0^{2\pi} r^2(\theta)\, d\theta .$$

The length is given by

$$L = \int_A^B ds = \int_{t_A}^{t_B} \sqrt{x'^2 + y'^2}\, dt .$$

Thus our function $h(t, x, y, x', y')$ is

$$h = \tfrac{1}{2}(xy' - x'y) + \lambda\sqrt{x'^2 + y'^2} .$$

We have one independent variable, t, two dependent variables, $x(t)$ and $y(t)$, and one constraint integral (and hence one Lagrange multiplier). Using Eq. (2.7),

we get

$$\frac{\partial h}{\partial x} - \frac{d}{dt}\left(\frac{\partial h}{\partial x'}\right) = 0 = \tfrac{1}{2}y' - \frac{d}{dt}\left(-\tfrac{1}{2}y + \lambda\frac{x'}{\sqrt{x'^2 + y'^2}}\right),$$

so that

$$y' - \lambda\frac{d}{dt}\left(\frac{x'}{\sqrt{x'^2 + y'^2}}\right) = 0 .$$

Similarly, the y equation yields

$$x' + \lambda\frac{d}{dt}\left(\frac{y'}{\sqrt{x'^2 + y'^2}}\right) = 0 .$$

Now these two equations are exact differentials so we may immediately integrate them to obtain

$$y - \lambda\frac{x'}{\sqrt{x'^2 + y'^2}} = y_0 , \qquad x + \lambda\frac{y'}{\sqrt{x'^2 + y'^2}} = x_0 ,$$

where x_0 and y_0 are constants to be determined by the fixed endpoint condition. We may rewrite these equations as

$$x - x_0 = -\lambda\frac{y'}{\sqrt{x'^2 + y'^2}} , \qquad y - y_0 = \lambda\frac{x'}{\sqrt{x'^2 + y'^2}} .$$

If we square both sides and add the two equations, we get immediately

$$(x - x_0)^2 + (y - y_0)^2 = \lambda^2 ,$$

so the extremal curve is a circle of radius λ about the point (x_0, y_0). Since the fixed length of the curve is L, we see that

$$\lambda = \frac{L}{2\pi} .$$

The points x_0 and y_0 are to be chosen so that the circle of radius $L/2\pi$ passes through the desired endpoint (which is the same as the starting point since the curve is closed). It is clear on physical grounds that the extremum here is a maximum, not a minimum.

The solution to this isoperimetric problem is particularly simple; it turns out to be just a circle. However, the reader should note that the equations obtained from the Euler-Lagrange formalism are very complicated, coupled, nonlinear differential equations. If one were to choose the functions $f(t, x, y, x', y')$ and $g(t, x, y, x', y')$ at random, the chances of being able to solve the resulting Euler-Lagrange equations in closed form would be small indeed.

Another famous isoperimetric problem is the determination of the shape of a hanging perfectly flexible rope of given length, fastened at its two endpoints. We assume the rope takes on a shape, consistent with the constraints, that minimizes its gravitational potential energy. Let the rope be held at its endpoints (x_A, y_A) and (x_B, y_B) as shown in Fig. 2.5. We want to determine the curve $y(x)$ passing through these fixed endpoints that minimizes the potential energy.

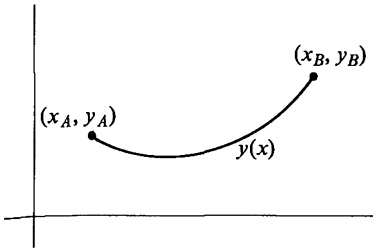

Fig. 2.5 A freely hanging rope held at its end points.

If ρ denotes the constant mass per unit length of the rope, then the potential energy relative to the x-axis is given by

$$I = \rho g \int_A^B y\, ds = \rho g \int_{x_A}^{x_B} y\sqrt{1 + y'^2}\, dx\,.$$

The constraint is that the length L of the rope is fixed:

$$L = \int_{x_A}^{x_B} ds = \int_{x_A}^{x_B} \sqrt{1 + y'^2}\, dx\,.$$

Thus

$$h = \rho g y\sqrt{1 + y'^2} - \rho g y_0 \sqrt{1 + y'^2}\,,$$

where we have written our Lagrange multiplier as $\lambda = -\rho g y_0$, y_0 to be determined. If we now change variables from y to $\eta = y - y_0$, we see that

$$h = \rho g \eta \sqrt{1 + \eta'^2}\,, \qquad \eta = \eta(x)\,.$$

But this has the same form as did the function f which we used to determine the minimum surface of revolution. Thus, just as in that case, the Euler-Lagrange equation will lead to

$$\eta = b \cosh\left(\frac{x - x_0}{b}\right),$$

where b and x_0 are constants to be determined. Since $\eta = y - y_0$, the equation of the curve having the shape of a freely hanging rope with fixed endpoints turns out to be

$$y = y_0 + b \cosh\left(\frac{x - x_0}{b}\right).$$

The three constants x_0, y_0, and b are chosen so that the curve passes through (x_A, y_A) and (x_B, y_B) and so that the length is L. This presents numerical problems, but is always possible in principle.

It may have occurred to the reader that there are other types of constraint equations which could arise in practice. We close this section by discussing

a particularly important class of constraints which make their appearance primarily in classical mechanics. Suppose that we have an integral

$$I = \int_{x_A}^{x_B} f(x, y_1, y_2, \cdots, y_n; y_1', y_2', \cdots, y_n')\, dx$$

which we want to extremize subject to the conditions

$$\begin{aligned} g_1(x, y_1, y_2, \cdots, y_n) &= 0\,, \\ g_2(x, y_1, y_2, \cdots, y_n) &= 0\,, \\ &\vdots \\ g_m(x, y_1, y_2, \cdots, y_n) &= 0\,, \end{aligned}$$

where $m < n$. (Otherwise we would have constrained ourselves out of business.) Now, one possible way to proceed would be to solve these m equations for m of the y's in terms of the remaining $(n - m)y$'s and insert these values into the integrand of I. However, in many cases the functions g_i are sufficiently complicated to make this an undesirable approach. Also, for reasons of symmetry it is often not desirable to single out some variables and write all the other variables in terms of these special ones. Finally, even if it is a fairly simple matter to eliminate variables, we might not want to do so for physical reasons. We shall see in the next section that this is often the case in classical mechanics.

We now outline an alternative procedure, treating the case in which we have just one constraint equation,

$$g(x, y_1, y_2 \cdots, y_n) = 0\,,$$

and want to minimize

$$I = \int_{x_A}^{x_B} f(x, y_1, y_2, \cdots, y_n; y_1', y_2', \cdots, y_n')\, dx\,.$$

As usual, we introduce a collection of test functions $\bar{y}_i(x, \epsilon_1, \epsilon_2)$ satisfying

α') $\bar{y}_i(x_A, \epsilon_1, \epsilon_2) = y_{iA}$, $\bar{y}_i(x_B, \epsilon_1, \epsilon_2) = y_{iB}$ for all ϵ_1, ϵ_2, that is, fixed endpoints again;

β') $\bar{y}_i(x, 0, 0) = y_i(x)$, where $y_i(x)$ are the desired extremizing functions; and

γ') $\bar{y}_i(x, \epsilon_1, \epsilon_2)$ has continuous derivatives through second order.

We now form

$$I(\epsilon_1, \epsilon_2) = \int_{x_A}^{x_B} f(x, \bar{y}_1, \cdots, \bar{y}_n; \bar{y}_1', \cdots, \bar{y}_n')\, dx\,,$$

and as before we require that $I(\epsilon_1, \epsilon_2)$ have an extremal value at $\epsilon_1 = 0 = \epsilon_2$, subject to

$$g(x, \bar{y}_1, \cdots, \bar{y}_n) = 0\,.$$

An equation of this form yields a *local* constraint, holding at all points, whereas constraints of the integral type encountered thus far are *global* con-

straints, whose implications are weaker than those of the local type. [The local equation actually implies that one of the $\bar{y}_i$ can be written in terms of the other $(n-1)$ $\bar{y}_i$, whereas the global constraint only gives a relation between the ϵ variables.] It would, of course, be possible to convert this equation into a global constraint just by integrating it from x_A to x_B, but in doing so we would lose most of the equation's content. However, we can retain the equation's generality and at the same time cast it into the familiar integral form if we first multiply it by an *arbitrary* function of x, say $\phi(x)$, and *then* integrate from x_A to x_B. Doing this, we obtain

$$J(\epsilon_1, \epsilon_2) = \int_{x_A}^{x_B} \phi(x) g(x, \bar{y}_1, \cdots, \bar{y}_n)\, dx = 0 ,$$

where $\phi(x)$ is some completely arbitrary function.

Just as we did in the first part of this section, we set

$$K(\epsilon_1, \epsilon_2) = I(\epsilon_1, \epsilon_2) + \lambda J(\epsilon_1, \epsilon_2)$$

and obtain a necessary condition for an extremum by requiring that

$$\left[\frac{\partial K}{\partial \epsilon_1}\right]_{\substack{\epsilon_1=0\\ \epsilon_2=0}} = 0 = \left[\frac{\partial K}{\partial \epsilon_2}\right]_{\substack{\epsilon_1=0\\ \epsilon_2=0}} , \tag{2.9}$$

where K is given by

$$K(\epsilon_1, \epsilon_2) = \int_{x_A}^{x_B} h(x, \bar{y}_1, \cdots, \bar{y}_n; \bar{y}_1', \cdots, \bar{y}_n')\, dx ,$$

and

$$h = f + \lambda\phi(x) g .$$

For notational convenience, we define $\lambda(x) \equiv \lambda\phi(x)$; $\lambda(x)$ is a completely undetermined function at present. Using our (by now) standard method, we compute the two first partial derivatives of $K(\epsilon_1, \epsilon_2)$ and find

$$\frac{\partial K}{\partial \epsilon_j} = \sum_{i=1}^{n} \int_{x_A}^{x_B} \left[\frac{\partial h}{\partial \bar{y}_i}\frac{\partial \bar{y}_i}{\partial \epsilon_j} + \frac{\partial h}{\partial \bar{y}_i'}\frac{\partial \bar{y}_i'}{\partial \epsilon_j}\right] dx , \qquad j = 1, 2 .$$

Integrating the second term by parts, we find that

$$\frac{\partial K}{\partial \epsilon_j} = \sum_{i=1}^{n} \int_{x_A}^{x_B} \left[\frac{\partial h}{\partial \bar{y}_i} - \frac{d}{dx}\left(\frac{\partial h}{\partial \bar{y}_i'}\right)\right]\frac{\partial \bar{y}_i}{\partial \epsilon_j}\, dx , \qquad j = 1, 2 ,$$

where condition (α') has enabled us to drop the terms coming from the endpoints. Now we impose Eq. (2.9) as a necessary condition for an extremum. Setting

$$\left[\frac{\partial \bar{y}_i}{\partial \epsilon_j}\right]_{\substack{\epsilon_1=0\\ \epsilon_2=0}} = \eta_j^{(i)}(x) , \tag{2.10}$$

we get

$$\sum_{i=1}^{n} \int_{x_A}^{x_B} \left[\frac{\partial h}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial h}{\partial y_i'}\right)\right]\eta_j^{(i)}(x)\, dx = 0 \tag{2.11}$$

for $j = 1, 2$. It is at this point that we must be careful. One is tempted to use the familiar argument to the effect that the functions $\eta_j^{(i)}(x)$ are completely arbitrary except for the fact that they are smooth and vanish at x_A and x_B. However, in this case, the argument is not correct! Let us take $g(x, y_1 \cdots, y_n) = 0$ and differentiate with respect to ϵ_1. We obtain

$$\frac{dg}{d\epsilon_1} = 0 = \frac{\partial g}{\partial \bar{y}_1}\frac{\partial \bar{y}_1}{\partial \epsilon_1} + \frac{\partial g}{\partial \bar{y}_2}\frac{\partial \bar{y}_2}{\partial \epsilon_1} + \cdots + \frac{\partial g}{\partial \bar{y}_n}\frac{\partial \bar{y}_n}{\partial \epsilon_1} .$$

If we now set $\epsilon_1 = 0 = \epsilon_2$ and use the definition of Eq. (2.10), we get

$$\sum_{i=1}^{n} \frac{\partial g}{\partial y_i}\eta_1^{(i)}(x) = 0 .$$

Thus there is a linear relation between the functions $\eta_1^{(i)}(x)$ and similarly between the functions $\eta_2^{(i)}(x)$, so we certainly cannot claim that the $\eta_j^{(i)}(x)$ are completely arbitrary. However, we still have the function $\lambda(x)$ at our disposal. Let us focus our attention on Eq. (2.11) with $j = 1$; the case $j = 2$ is handled analogously. Since g does not depend on the derivatives of the $y_i(x)$, we may write Eq. (2.11) in more detail as

$$\sum_{i=1}^{n} \int_{x_A}^{x_B} \left[\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial y_i'}\right) + \lambda(x)\frac{\partial g}{\partial y_i}\right]\eta_1^{(i)}(x)\, dx = 0 . \tag{2.12}$$

Now let us assume that $\eta_1^{(1)}(x)$ is given in terms of the remaining $\eta_1^{(i)}(x)$. Rather than substitute this value of $\eta_1^{(1)}(x)$ into Eq. (2.12), let us choose $\lambda(x)$ so that

$$\frac{\partial f}{\partial y_1} - \frac{d}{dx}\left(\frac{\partial f}{\partial y_1'}\right) + \lambda(x)\frac{\partial g}{\partial y_1} = 0 . \tag{2.13}$$

Thus we need not know the expression for $\eta_1^{(1)}(x)$, since its coefficient in our basic equation is zero. With $\lambda(x)$ determined in this way, Eq. (2.12) becomes

$$\sum_{i=2}^{n} \int_{x_A}^{x_B} \left[\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial y_i'}\right) + \lambda(x)\frac{\partial g}{\partial y_i}\right]\eta_1^{(i)}(x)\, dx = 0 ,$$

where the term $i = 1$ no longer appears. Now the last $n - 1$ of the $\eta_1^{(i)}(x)$ are completely independent, so we may safely conclude that

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial y_i'}\right) + \lambda(x)\frac{\partial g}{\partial y_i} = 0 , \qquad i = 2, 3, \cdots, n . \tag{2.14}$$

These $n - 1$ equations are identical in form with Eq. (2.13), but we obtained them in a rather different manner. We can now combine Eqs. (2.13) and (2.14) into the single relation

$$\frac{\partial h}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial h}{\partial y_i'}\right) = 0 , \qquad i = 1, 2, \cdots, n , \tag{2.15}$$

where $h = f + \lambda(x)g$. Note that we need not have singled out $\eta^{(1)}(x)$ in obtaining Eq. (2.15); any $\eta^{(i)}(x)$ would have led to the same result in which all the original dependent variables appear on an equal footing.

Thus we have n second-order differential equations for the n unknown functions $y_n(x)$ and the function $\lambda(x)$. [Note that no derivatives of $\lambda(x)$ appear. This is because we have assumed that g does *not* depend on the derivatives of the $y_i(x)$. Hence these equations are not differential equations as far as $\lambda(x)$ is concerned.] The additional equation necessary to make the problem well-posed is just

$$g(x, y_1, y_2, \cdots, y_n) = 0 ,$$

where g is a *given* function. The solutions $y_1(x), y_2(x), \cdots, y_n(x)$ are determined uniquely by the boundary conditions

$$y_i(x_A) = y_{iA} , \qquad y_i(x_B) = y_{iB} , \qquad i = 1, 2, \cdots, n .$$

Finally, the reader should check that if we have m constraint equations of the form

$$g_i(x, y_1, y_2, \cdots, y_n) = 0 , \qquad i = 1, 2, \cdots, m ,$$

then we again obtain the system of equations given by Eq. (2.15), but in this case the function h is given by

$$h = f + \sum_{i=1}^{m} \lambda_i(x) g_i ,$$

where the $\lambda_i(x)$ are to be determined as part of the solution to the system.

2.5 APPLICATION TO CLASSICAL MECHANICS

With the mathematical equipment of the last four sections, a fairly large part of classical mechanics can be conveniently expressed. If a classical-mechanical system, specified by generalized coordinates $q_1(t), q_2(t), \cdots, q_n(t)$, has a potential function $V(q_1, q_2, \cdots, q_n, t)$, then its Lagrangian can be constructed:

$$L(t, q_i, \dot{q}_i) = T(q_i, \dot{q}_i) - V(q_i, t) ,$$

where T is the kinetic energy, and we use a dot to denote differentiation with respect to time. It can be shown that the motion of the system now follows from Hamilton's principle.

Hamilton's principle. The motion of system whose Lagrangian is L, from time t_A to time $t_B (t_A < t_B)$, is such as to render the integral

$$I = \int_{t_A}^{t_B} L \, dt$$

an extremum with respect to the functions $q_i(t)$ $(i = 1, 2, \cdots, n)$ for which $q_i(t_A)$ and $q_i(t_B)$ are prescribed; t_A and t_B are arbitrary, but fixed, times.

It follows immediately from our work that I is extremized if and only if

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = 0 \qquad i = 1, 2, \cdots, n . \tag{2.16}$$

These are Lagrange's equations of motion, a set of n simultaneous second-order differential equations whose solution gives the functions $q_i(t)$. The $2n$ constants of integration are evaluated by specifying the state of the system at some of time, t_A, that is by giving the values of $q_i(t_A)$ and $\dot{q}_i(t_A)$ for all i.

For example, suppose that

$$T = \tfrac{1}{2} m(\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2) , \qquad V = V(x_1, x_2, x_3) ,$$

that is, suppose that a particle of mass m is moving in an arbitrary potential field. Then

$$L = \tfrac{1}{2} m(\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2) - V(x_1, x_2, x_3) ,$$

and Lagrange's equations give us

$$-\frac{\partial V}{\partial x_i} - \frac{d}{dt}(m\dot{x}_i) = 0 , \qquad i = 1, 2, 3 ,$$

or

$$m\ddot{x}_i = -\frac{\partial V}{\partial x_i} .$$

Since $-\partial V/\partial x_i$ is the force on the particle in the x_i-direction, this is simply Newton's second law which may be written in vector form as

$$m\ddot{\mathbf{r}} = -\nabla V = \mathbf{F} .$$

We should point out that it is possible to extend the Lagrangian formulation to more general potentials, such as the velocity-dependent potentials of electromagnetic theory, if the forces can be written as

$$F_i = -\frac{\partial V}{\partial q_i} + \frac{d}{dt}\left(\frac{\partial V}{\partial \dot{q}_i}\right).$$

In this case, we again find that

$$L = T - V ,$$

and Hamilton's principle holds. Thus Eq. (2.16) will still govern the motion of such a system.

If L does not depend explicitly on t, we have immediately from Eq. (2.6)

$$\sum_{i=1}^{n} \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i - L = E , \tag{2.17}$$

where E is a constant. The significance of E may be seen as follows. If V does not depend on the $\dot{q}_i$, then

$$\sum_{i=1}^{n} \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i = \sum_{i=1}^{n} \frac{\partial T}{\partial \dot{q}_i}\dot{q}_i .$$

Under many circumstances T will take the form

$$T = \sum_{l,m} a_{lm}(q_i)\dot{q}_l\dot{q}_m$$

so that

$$\sum_{i=1}^{n} \frac{\partial T}{\partial \dot{q}_i} \dot{q}_i = 2T .$$

Thus Eq. (2.17) can be written as

$$2T - L = 2T - (T - V) = T + V = E ,$$

that is, we may identify E with the total ("conserved") energy.

The quantity

$$p_i \equiv \frac{\partial L}{\partial \dot{q}_i}$$

is called the *generalized momentum* corresponding to (or *conjugate* to) the generalized coordinate q_i. If some particular $q_k(t)$ does not appear in L, $q_k(t)$ is said to be an *ignorable* coordinate. For the generalized momentum p_k conjugate to an ignorable coordinate, we have

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \dot{p}_k = 0$$

so that p_k is a constant. This is just a statement of the conservation of momentum.

As an example, let us consider the motion of a particle in a plane, when the potential (and hence the force) depends only on the radial coordinate. Using polar coordinates in the plane, we have

$$L = \tfrac{1}{2} m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r) ;$$

θ is an ignorable coordinate, and therefore

$$\frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} = p_\theta , \tag{2.18}$$

where p_θ is a constant which is seen to be the angular momentum of the particle. In this case, since the Lagrangian does not depend explicitly on t, we also have from Eq. (2.17),

$$\tfrac{1}{2} m(\dot{r}^2 + r^2\dot{\theta}) + V(r) = E . \tag{2.19}$$

Combining Eqs. (2.18) and (2.19), we obtain

$$\dot{\theta} = \frac{p_\theta}{mr^2} , \qquad \tfrac{1}{2} m\left(\dot{r}^2 + \frac{p_\theta^2}{m^2r^2}\right) + V(r) = E ,$$

so in this case our constants of the motion, p_θ and E, reduce the problem to one involving only first derivatives, i.e., the problem is reduced to evaluating some integrals. Of course, for a complicated $V(r)$ these integrals might have to be done numerically.

Finally, let us consider a mechanical problem with a constraint. In particular we ask: What is the motion of a particle moving under gravity if it is constrained to move always on the surface of a fixed solid sphere of radius

a? It will clearly be to our advantage to use spherical coordinates. Therefore we write

$$T = \tfrac{1}{2} m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 \sin^2\theta\dot{\phi}^2) ,$$
$$V = mgz = mgr\cos\theta ,$$

and

$$L = \tfrac{1}{2} m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 \sin^2\theta\dot{\phi}^2) - mgr\cos\theta .$$

We want to extremize

$$I = \int_{t_A}^{t_B} L\, dt ,$$

subject to the constraint

$$g(t, r, \theta, \phi) = r - a = 0 .$$

According to our discussion in Section 2.4, we form the function

$$\begin{aligned} L' &= L + \lambda g \\ &= \tfrac{1}{2} m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 \sin^2\theta\dot{\phi}^2) - mgr\cos\theta + \lambda r - \lambda a \end{aligned}$$

and use this function in the Euler-Lagrange equations. For the r-coordinate, we find that

$$\lambda - mg\cos\theta + mr\sin^2\theta\dot{\phi}^2 + mr\dot{\theta}^2 - m\ddot{r} = 0 , \tag{2.20a}$$

and for the θ-coordinate,

$$mgr\sin\theta + mr^2\sin\theta\cos\theta\dot{\phi}^2 - m\frac{d}{dt}(r^2\dot{\theta}) = 0 . \tag{2.20b}$$

Now ϕ is an ignorable coordinate, so we get immediately

$$mr^2 \sin^2\theta\dot{\phi} = p_\phi$$

from which we obtain

$$\dot{\phi} = \frac{p_\phi}{mr^2\sin^2\theta} . \tag{2.20c}$$

Thus we have three coupled differential equations to solve, together with the relation

$$r - a = 0 . \tag{2.20d}$$

Putting Eqs. (2.20c) and (2.20d) into Eqs. (2.20a) and (2.20b), we get

$$\lambda = mg\cos\theta - ma\dot{\theta}^2 - \frac{p_\phi^2}{ma^3\sin^2\theta} \tag{2.21a}$$

and

$$\ddot{\theta} = \frac{g}{a}\sin\theta + \frac{p_\phi^2\cos\theta}{m^2a^4\sin^3\theta} . \tag{2.21b}$$

Eq. (2.20c) becomes simply

$$\dot{\phi} = \frac{p_\phi}{ma^2 \sin^2 \theta} . \tag{2.21c}$$

Thus we see that everything hinges on the solution to Eq. (2.21b); once $\theta(t)$ is determined, Eq. (2.21a) gives us $\lambda(t)$ directly, and upon integrating Eq. (2.21c) we will get $\phi(t)$. In fact, Eq. (2.21b) is not as complicated as it looks. If we multiply both sides of the equation by $\dot{\theta}(t)$ they both become exact differentials, and we can integrate once to obtain

$$\dot{\theta}^2 = C - 2\frac{g}{a}\cos\theta - \frac{p_\phi^2}{m^2a^4 \sin^2\theta} . \tag{2.22}$$

Such a relation is not surprising; the reader can show that this follows directly from the conservation of energy. The constant C is just equal to $2E/ma^2$, where E is the total conserved energy. Hence the problem is reduced to evaluating the integral implied by Eq. (2.22). This integral cannot, in general, be evaluated in closed form, although in certain limiting cases general features of the solution can be determined without much difficulty.

It is interesting to consider $\lambda(t)$ in some detail. The first term in Eq. (2.21a) for $\lambda(t)$ is the negative of the radial component of the gravitational force on the particle. The second term is just the negative of the centrifugal force due to the θ-motion (a radial force), and the last term is the radial component of the centrifugal force due to the ϕ-motion. Thus $\lambda(t)$ is precisely the force which the sphere must exert to keep the particle on the sphere. One can show that in general the Lagrange multipliers are related in a simple way to the forces required to maintain the constraints in question. Thus they provide us with direct physical information about the system under study.

2.6 EXTREMIZATION OF MULTIPLE INTEGRALS

As our final extension of the variational problem, we now want to consider the case when the integral to be extremized has more than one independent variable. We shall deal directly with the case of two independent variables and leave the more general case as an exercise.

Consider, then, the integral

$$I = \iint_D f\left(x_1, x_2, y, \frac{\partial y}{\partial x_1}, \frac{\partial y}{\partial x_2}\right) dx_1\, dx_2 ,$$

where x_1 and x_2 are the two independent variables and D is some fixed domain in the x_1x_2-plane. We shall denote the boundary of D by C. Just as before, we shall prescribe the values of $y(x_1, x_2)$ at all points of the boundary C, that is,

$$[y(x_1, x_2)]_{(x_1,x_2)\in C} = g(C) ,$$

where $g(C)$ is a given function, defined on the one-dimensional curve, C.

Therefore we want our one-parameter family of test functions to satisfy

a) $[\bar{y}(x_1, x_2, \epsilon)]_{(x_1, x_2)\epsilon C} = g(C)$ for all values of ϵ;

b) $\bar{y}(x_1, x_2, 0) = y(x_1, x_2)$, the desired extremizing function; and

c) $\bar{y}(x_1, x_2, \epsilon)$ has continuous first and second derivatives.

We now form

$$I(\epsilon) = \iint_D f\left(x_1, x_2, \bar{y}, \frac{\partial \bar{y}}{\partial x_1}, \frac{\partial \bar{y}}{\partial x_2}\right) dx_1\, dx_2$$

and require that

$$\left[\frac{dI}{d\epsilon}\right]_{\epsilon=0} = 0 .$$

Proceeding in the usual manner, we find that

$$\frac{dI}{d\epsilon} = \iint_D \left[\frac{\partial f}{\partial \bar{y}}\frac{d\bar{y}}{d\epsilon} + \frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_1}\right)}\frac{d}{d\epsilon}\left(\frac{\partial \bar{y}}{\partial x_1}\right) + \frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_2}\right)}\frac{d}{d\epsilon}\left(\frac{\partial \bar{y}}{\partial x_2}\right)\right] dx_1\, dx_2$$

$$= \iint_D \left[\frac{\partial f}{\partial \bar{y}}\frac{d\bar{y}}{d\epsilon} + \frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_1}\right)}\frac{\partial}{\partial x_1}\left(\frac{d\bar{y}}{d\epsilon}\right) + \frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_2}\right)}\frac{\partial}{\partial x_2}\left(\frac{d\bar{y}}{d\epsilon}\right)\right] dx_1\, dx_2 .$$

We now need to make use of the two-dimensional generalization of integration by parts—Green's Theorem—which reads

$$\iint_D \left[\frac{\partial P}{\partial x_1} + \frac{\partial Q}{\partial x_2}\right] dx_1\, dx_2 = \int_C (P\, dx_2 - Q\, dx_1) .$$

This is valid if $P(x_1, x_2)$ and $Q(x_1, x_2)$ and their first partial derivatives are continuous in D. If we write

$$P = \phi(x_1, x_2)A(x_1, x_2) \quad \text{and} \quad Q = \phi(x_1, x_2)B(x_1, x_2),$$

then we get

$$\iint_D \left[A\frac{\partial \phi}{\partial x_1} + B\frac{\partial \phi}{\partial x_2}\right] dx_1\, dx_2 = -\iint_D \left[\frac{\partial A}{\partial x_1} + \frac{\partial B}{\partial x_2}\right]\phi\, dx_1\, dx_2$$
$$+ \int_C [A\, dx_2 - B\, dx_1]\,\phi .$$

By making the identification

$$\phi = \frac{d\bar{y}}{d\epsilon}, \qquad A = \frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_1}\right)}, \qquad B = \frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_2}\right)},$$

we obtain

$$\frac{dI}{d\epsilon} = \iint_D \left[\frac{\partial f}{\partial \bar{y}} - \frac{\partial}{\partial x_1}\frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_1}\right)} - \frac{\partial}{\partial x_2}\frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_2}\right)}\right]\frac{d\bar{y}}{d\epsilon}\, dx_1\, dx_2$$
$$+ \int_C \frac{d\bar{y}}{d\epsilon}\left[\frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_1}\right)} dx_2 - \frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_2}\right)} dx_1\right].$$

But by condition (a), $d\bar{y}/d\epsilon$ vanishes on C, so we get, using our extremum requirement,

$$\iint_D \left[\frac{\partial f}{\partial \bar{y}} - \frac{\partial}{\partial x_1}\frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_1}\right)} - \frac{\partial}{\partial x_2}\frac{\partial f}{\partial\left(\frac{\partial \bar{y}}{\partial x_2}\right)}\right]\eta(x_1, x_2)\, dx_1\, dx_2 = 0\,,$$

where

$$\left[\frac{d\bar{y}}{d\epsilon}\right]_{\epsilon=0} = \eta(x_1, x_2)\,.$$

Since $\eta(x_1, x_2)$ is an *arbitrary* smooth function inside D, the equation

$$\frac{\partial f}{\partial y} - \frac{\partial}{\partial x_1}\frac{\partial f}{\partial\left(\frac{\partial y}{\partial x_1}\right)} - \frac{\partial}{\partial x_2}\frac{\partial f}{\partial\left(\frac{\partial y}{\partial x_2}\right)} = 0 \tag{2.23}$$

must hold everywhere in D. This is the Euler-Lagrange equation for two independent variables; the partial derivatives reflect the presence of more than one independent variable.

As an example of a variational problem with two independent variables, we consider the vibrating string. Let a perfectly flexible elastic string be stretched under constant tension τ along the x-axis with its endpoints fixed at $x = 0$ and $x = l$. We consider small, transverse, frictionless (undamped) vibrations—the simplest possible case—and we neglect gravity. The amplitude of the vibration, at time t and position x $(0 \leq x \leq l)$, is $y(x, t)$. The slope of the string is given by $\partial y/\partial x$ and assumed to be small:

$$\left|\frac{\partial y}{\partial x}\right| \ll 1\,.$$

This will be the case in most situations that arise in practice. The velocity is $\partial y/\partial t$. The fixed endpoint condition says that $y(0, t) = y(l, t) = 0$ for all t.

We now want to compute the Lagrangian for this mechanical system and then determine the motion by Hamilton's principle. First we find the potential energy V. The work done on the string to effect a given distorted configuration all goes into changing the length of the string because we assume perfect

flexibility. Since the stretching force is equal to the (constant) tension τ, the potential energy is simply

$$V = \tau\left[\int_0^l \sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2}\, dx - l\right].$$

The term in brackets is the difference between the instantaneous length of the string and its equilibrium length l. The zero of V thus occurs when the length of the string is equal to l. Since we are assuming

$$\left|\frac{\partial y}{\partial x}\right| \ll 1,$$

we may write

$$\sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2} \cong 1 + \frac{1}{2}\left(\frac{\partial y}{\partial x}\right)^2.$$

To compute the kinetic energy, we assume a mass density along the string $\rho(x) > 0$. Thus the mass contained in an element of length dx is $\rho(x)\, dx$, with the associated kinetic energy

$$dT = \tfrac{1}{2}\rho(x)\, dx \left(\frac{\partial y}{\partial t}\right)^2.$$

The total kinetic energy is therefore

$$T = \tfrac{1}{2}\int_0^l \rho(x)\left(\frac{\partial y}{\partial t}\right)^2 dx.$$

With T and V determined, we now form the Lagrangian:

$$L = T - V = \int_0^l \left[\tfrac{1}{2}\rho(x)\left(\frac{\partial y}{\partial t}\right)^2 - \tfrac{1}{2}\tau\left(\frac{\partial y}{\partial x}\right)^2\right] dx.$$

The quantity in brackets,

$$\mathscr{L} = \tfrac{1}{2}\rho(x)\left(\frac{\partial y}{\partial t}\right)^2 - \tfrac{1}{2}\tau\left(\frac{\partial y}{\partial x}\right)^2,$$

is called the *Lagrangian density*, since its space integral yields the Lagrangian. Now, Hamilton's principle says that the function which describes the actual motion of the string is the one which renders

$$I = \int_{t_1}^{t_2} L\, dt = \int_{t_1}^{t_2}\int_0^l \mathscr{L}\, dx\, dt$$

an extremum with respect to those functions $y(x, t)$ which satisfy $y(0, t) = y(l, t) = 0$ for all t and for which $y(x, t_1)$ and $y(x, t_2)$ are given functions of x.

We now have available the mathematical tools necessary for determining the equation which $y(x, t)$ must satisfy. The domain D in the xt-plane is a rectangle defined by $0 \le x \le l$ and $t_1 \le t \le t_2$. Using Eq. (2.23), we have

$$\frac{\partial \mathscr{L}}{\partial y} - \frac{\partial}{\partial x}\frac{\partial \mathscr{L}}{\partial\left(\frac{\partial y}{\partial x}\right)} - \frac{\partial}{\partial t}\frac{\partial \mathscr{L}}{\partial\left(\frac{\partial y}{\partial t}\right)} = 0,$$

and inserting our expression for $\mathscr{L}$, we get

$$\frac{\partial^2 y}{\partial x^2} = \frac{\rho(x)}{\tau}\frac{\partial^2 y}{\partial t^2} .$$

This is the partial differential equation which describes the motion of the string —the wave equation.

We now generalize to the case of n independent variables. We want to extremize the multiple integral

$$I = \int_D \mathscr{L}\left(x_1, x_2, \cdots, x_n, \phi; \frac{\partial\phi}{\partial x_1}, \frac{\partial\phi}{\partial x_2}, \cdots, \frac{\partial\phi}{\partial x_n}\right) dx_1\, dx_2 \cdots dx_n ,$$

where $\phi = \phi(x_1, x_2 \cdots, x_n)$, D is some domain in the n-dimensional space, and ϕ is some prescribed function defined on the $(n-1)$-dimensional boundary of D. The Euler-Lagrange equation which gives the extremizing function ϕ is

$$\frac{\delta\mathscr{L}}{\delta\phi} \equiv \frac{\partial\mathscr{L}}{\partial\phi} - \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\frac{\partial\mathscr{L}}{\partial\left(\dfrac{\partial\phi}{\partial x_i}\right)} = 0 , \tag{2.24}$$

where we have introduced $\delta\mathscr{L}/\delta\phi$, the so-called functional derivative. We have already seen that the Lagrangian density

$$\mathscr{L} = \tfrac{1}{2}\left[\rho(x)\left(\frac{\partial\phi}{\partial t}\right)^2 - \tau\left(\frac{\partial\phi}{\partial x}\right)^2\right]$$

leads to the one-dimensional wave equation.

As another example we consider the Lagrangian density

$$\begin{aligned}\mathscr{L} &= (\nabla\phi)^2 = \nabla\phi\cdot\nabla\phi \\ &= \left[\left(\frac{\partial\phi}{\partial x}\right)^2 + \left(\frac{\partial\phi}{\partial y}\right)^2 + \left(\frac{\partial\phi}{\partial z}\right)^2\right].\end{aligned}$$

Note that from the basic properties of the gradient, $\mathscr{L}$ is invariant under rotations. The Euler-Lagrange equation for the case of three independent variables and one dependent variable is

$$\frac{\partial\mathscr{L}}{\partial\phi} - \sum_{i=1}^{3} \frac{\partial}{\partial x_i}\frac{\partial\mathscr{L}}{\partial\left(\dfrac{\partial\phi}{\partial x_i}\right)} = 0 ,$$

where $x_1 = x$, $x_2 = y$, $x_3 = z$. Substituting $\mathscr{L} = (\nabla\phi)^2$, we get

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = \nabla^2\phi = 0 .$$

Thus Laplace's equation is the necessary condition for extremizing

$$I = \int_D (\nabla\phi)^2\, dx\, dy\, dz .$$

If $\phi(x, y, z)$ is fixed on the boundary of D. We see that a rotationally invariant $\mathscr{L}$ has led to a rotationally invariant equation for ϕ.

These results may be further generalized to the case where there are several, say m, dependent variables, $\phi_j(x_1, x_2, \cdots, x_n)$, involved in $\mathscr{L}$. For example, if we extremize

$$I = \int_D \mathscr{L}\left(x_i, \phi_j, \frac{\partial \phi_j}{\partial x_i}\right) dx_1\, dx_2 \cdots dx_n ,$$

we obtain the set of m coupled second-order partial differential equations

$$\frac{\partial \mathscr{L}}{\partial \phi_j} - \sum_{i=1}^{n} \frac{\partial}{\partial x_i} \frac{\partial \mathscr{L}}{\partial\left(\frac{\partial \phi_j}{\partial x_i}\right)} = 0 , \qquad j = 1, 2, \cdots, m . \tag{2.25}$$

These versions of the Euler-Lagrange equation are used in field theory where the partial differential equations governing certain physical situations are derived by constructing a "reasonable" Lagrangian density $\mathscr{L}$ and then postulating a generalized Hamilton's principle which leads to the vanishing of the functional derivative, in other words, to the Euler-Lagrange equation. Reasonable Lagrangians may be partially determined from invariance considerations and symmetry properties: e.g., the Lagrangian density may be required to be Lorentz invariant, or invariant under certain gauge transformations.

As yet nothing has been said in this section about constraints that lead to isoperimetric problems. The generalization of this procedure to isoperimetric problems goes through exactly as before. We mention the result here for the general case of m dependent variables, n independent variables, and p integral constraints of the form

$$J_k = \int_D g_k\left(x_i, \phi_j, \frac{\partial \phi_j}{\partial x_i}\right) dx_1\, dx_2 \cdots dx_n ,$$

where $i = 1, \cdots, n$; $j = 1, \cdots, m$; $k = 1, \cdots, p$. Here the J_k are p fixed constants. Subject to these p constraints we want to extremize the multiple integral

$$I = \int_D \mathscr{L}\left(x_i, \phi_j, \frac{\partial \phi_j}{\partial x_i}\right) dx_1\, dx_2 \cdots dx_n .$$

In strict analogy with the isoperimetric problem discussed previously, we do this by forming the new function

$$h = \mathscr{L} + \sum_{k=1}^{p} \lambda_k g_k , \tag{2.26}$$

where the λ_k are p constant Lagrange multipliers. Now h satisfies the appropriate system of Euler-Lagrange equations, namely,

$$\frac{\partial h}{\partial \phi_i} - \sum_{i=1}^{n} \frac{\partial}{\partial x_i} \frac{\partial h}{\partial\left(\frac{\partial \phi_j}{\partial x_i}\right)} = 0, \qquad j = 1, \cdots, m . \tag{2.27}$$

As an illustration of this general formalism, let us consider the Lagrange density

$$\mathscr{L} = \frac{\hbar^2}{2m}\left[\frac{\partial\psi^*}{\partial x}\frac{\partial\psi}{\partial x} + \frac{\partial\psi^*}{\partial y}\frac{\partial\psi}{\partial y} + \frac{\partial\psi^*}{\partial z}\frac{\partial\psi}{\partial z}\right] + V(x, y, z)\psi^*\psi ,$$

where ψ^* (x, y, z) and $\psi(x, y, z)$ are complex-valued functions and V is real-valued. (An asterisk denotes complex conjugation.) We want to extremize

$$I = \int_D \mathscr{L}\, dx\, dy\, dz ,$$

subject to the constraint

$$J = \int_D \psi^*\psi\, dx\, dy\, dz = 1 .$$

This is a variational problem with three independent variables (x, y, z), two dependent variables (the real and imaginary parts of ψ, ψ_R and ψ_I respectively), and one constraint. According to Eq. (2.26), we now form

$$\begin{aligned} h &= \mathscr{L} + \lambda\psi^*\psi \\ &= \frac{\hbar^2}{2m}\left[\left(\frac{\partial\psi_R}{\partial x}\right)^2 + \left(\frac{\partial\psi_R}{\partial y}\right)^2 + \left(\frac{\partial\psi_R}{\partial z}\right)^2 + \left(\frac{\partial\psi_I}{\partial x}\right)^2 + \left(\frac{\partial\psi_I}{\partial y}\right)^2 + \left(\frac{\partial\psi_I}{\partial z}\right)^2\right] \\ &\quad + [V(x, y, z) + \lambda][\psi_R^2 + \psi_I^2] . \end{aligned}$$

Substituting into Eq. (2.27) and making the identification $\phi_1 = \psi_R$, $\phi_2 = \psi_I$, we get for $j = 1$,

$$\lambda\psi_R + V\psi_R - \frac{\hbar^2}{2m}\sum_{i=1}^{3}\frac{\partial}{\partial x_i}\left(\frac{\partial\psi_R}{\partial x_i}\right) = 0$$

and for $j = 2$,

$$\lambda\psi_I + V\psi_I - \frac{\hbar^2}{2m}\sum_{i=1}^{3}\frac{\partial}{\partial x_i}\left(\frac{\partial\psi_I}{\partial x_i}\right) = 0 ,$$

where as usual we call $x = x_1$, $y = x_2$, $z = x_3$. Calling $\lambda = -E$, we may write these as a single equation,

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi .$$

This equation is the time-independent Schrödinger equation for the motion of a nonrelativistic particle in a potential $V(x, y, z)$. In this case, the Lagrange multiplier plays the role of the energy eigenvalue in the Schrödinger equation. We shall see this in a more general setting in Chapter 4, where we shall also consider the interesting question of whether or not minima occur at the extreme values.

*2.7 INVARIANCE PRINCIPLES AND NOETHER'S THEOREM

Having introduced the basic concepts of the calculus of variations, we now look back to the starting point of our investigations and ask why it was, that in considering the trial function $\bar{y}(x, \epsilon)$ in previous sections, we always demanded that $\bar{y}(x, \epsilon)$ obey fixed endpoint conditions. Clearly, there is a large class of problems which falls into this category, but there are many others which lie outside it; for example, the problem of finding the path connecting a fixed point P to a vertical line L such that a particle moving under gravity will go from P to L in minimum time. Here one endpoint is fixed and one is free to lie anywhere on L.

From the mathematical point of view, the reason for starting with fixed endpoint restrictions is to look at the simplest things first. From the point of view of the physicist, the reason is clearer, namely, Hamilton's principle refers to variations of the fixed endpoint variety. Since this principle has played a crucial role in the development of physics, it is not surprising that the physicist's attention is first focussed on this special variational problem.

However, now that we have familiarized ourselves with the simplest problems, let us look at the problem of varying the integral

$$I = \int_{x_A}^{x_B} f(x, y, y')\, dx \tag{2.28}$$

from a more general point of view. For simplicity, we will consider the case of infinitesimal transformations, parametrized by a small number ϵ. If we were working in the context of previous sections, we would consider a transformation

$$y(x) \to \bar{y}(x) \equiv y(x) + \epsilon\eta(x) ,$$

which is just a special case of the transformation $y(x) \to \bar{y}(x, \epsilon)$ considered in Section 2.2, *if* we require that $\eta(x)$ vanish at x_A and x_B so that $\bar{y}(x_A) = y(x_A)$ and $\bar{y}(x_B) = y(x_B)$ for *all* ϵ. The simplest generalization of this type of variation consists of dropping the restrictions on $\eta(x)$. A further generalization would be to allow η to have a more complicated functional form. For example, η might depend explicitly on the functions y and y'; then we would write

$$y(x) \to \bar{y}(x, y, y') \equiv y(x) + \epsilon\eta(x, y, y') . \tag{2.29}$$

The primary reason for considering such elaborate transformations, particularly those for which the endpoints are not fixed, is that there are many transformations—often very simple ones—in the physical world which are of this type. For example, we might want to consider a translated trajectory of a classical particle $[x(t) \to x(t) + \epsilon]$ or perhaps a rotated trajectory. We will see that by allowing for the possibility of variations such as these we will be able to obtain very general information about quantities which are conserved in physical problems.

Pursuing the analogy suggested by classical physics (via Hamilton's principle) we can also imagine the possibility of translating a particle trajectory in

time, for example, by the transformation $t \to t + \epsilon$. In order to allow for variations of this type, we consider another extension of the variational approach, which is to *shift* (not necessarily in a uniform manner) the curve along which we integrate to some other region of space. For example, we could transform each point x into a point $\bar{x}$ via the transformation

$$x \to \bar{x} = x + \epsilon\xi(x, y, y') , \tag{2.30}$$

that is, this transformation depends not only on the point x in question, but also on the function y occurring in the integrand of I. With the transformations of Eqs. (2.29) and (2.30), we can form a quantity $\bar{I}(\epsilon)$ analogous to the I of Eq. (2.28):

$$\bar{I}(\epsilon) = \int_{\bar{x}_A}^{\bar{x}_B} f[\bar{x}, \bar{y}(\bar{x}), \bar{y}'(\bar{x})]\, d\bar{x} . \tag{2.31}$$

Note that $\bar{I}(\epsilon = 0) = I$. In Eq. (2.31), $\bar{x}_A$ and $\bar{x}_B$ are obtained by applying the transformation of Eq. (2.30) to x_A and x_B, respectively. In Eq. (2.31), the significant quantity is $\bar{y}(\bar{x})$. This symbol does *not* mean that we just replace x by $\bar{x}$ in Eq. (2.29). Rather, it means that we consider the *transformed curve* $\bar{y}$, obtained by writing x in terms of $\bar{x}$ [using Eq. (2.30)] and substituting that value of x into Eq. (2.29). Since we will be interested only in evaluating $\bar{I}(\epsilon)$ *through first order in* ϵ, we can use x or $\bar{x}$ interchangeably in the functions ξ and η, because in zeroth order, x and $\bar{x}$ are equal. In Eq. (2.31), the prime on $\bar{y}'$ denotes differentiation with respect to $\bar{x}$, while in Eq. (2.28) the prime on y' denotes differentiation with respect to x, that is, a prime will always denote differentiation with respect to the *independent* variable, whatever it may be.

Since ϵ is small, let us try to compute $\bar{I}(\epsilon)$ in power series form. For reasons that will soon become clear, we will concentrate on the first two terms in the series. Using Eqs. (2.29) and (2.30), we have

$$\bar{y}(\bar{x}) = y[(\bar{x} - \epsilon\xi(\bar{x}, y, y')] + \epsilon\eta(\bar{x}, y, y') ,$$

where y and y' are functions of $\bar{x}$. Expanding through terms in ϵ, we find that

$$\bar{y}(\bar{x}) = y(\bar{x}) - \epsilon\frac{\partial y}{\partial \bar{x}}\xi(\bar{x}, y, y') + \epsilon\eta(\bar{x}, y, y')$$

or

$$\bar{y}(\bar{x}) = y(\bar{x}) + \epsilon\rho(\bar{x}, y, y') , \tag{2.32}$$

where we have made the definition

$$\rho(\bar{x}, y, y') \equiv \eta(\bar{x}, y, y') - \frac{\partial y}{\partial \bar{x}}\xi(\bar{x}, y, y') \tag{2.33}$$

in order to simplify subsequent calculations. Using Eqs. (2.31) and (2.32), we can write $\bar{I}(\epsilon)$ as

$$\bar{I}(\epsilon) = \int_{\bar{x}_A}^{\bar{x}_B} f[\bar{x}, y(\bar{x}) + \epsilon\rho(\bar{x}, y, y'), y'(\bar{x}) + \epsilon\rho'(\bar{x}, y, y')]\, d\bar{x} . \tag{2.34}$$

Now, according to Eq. (2.30),

$$\bar{x}_A = x_A + \epsilon\xi(x, y, y')\,|_{x=x_A} \equiv x_A + \delta_A\,, \tag{2.35a}$$

$$\bar{x}_B = x_B + \epsilon\xi(x, y, y')\,|_{x=x_B} \equiv x_B + \delta_B\,, \tag{2.35b}$$

so let us break up the integral in Eq. (2.34) as follows:

$$\int_{\bar{x}_A}^{\bar{x}_B} = \int_{x_A+\delta_A}^{x_B+\delta_B} = \int_{x_A+\delta_A}^{x_B} + \int_{x_B}^{x_B+\delta_B} = \int_{x_A}^{x_B} - \int_{x_A}^{x_A+\delta_A} + \int_{x_B}^{x_B+\delta_B}$$

Thus

$$\begin{aligned}\bar{I}(\epsilon) = &\int_{x_A}^{x_B} f[\bar{x}, y(\bar{x}) + \epsilon\rho(\bar{x}, y, y'), y'(\bar{x}) + \epsilon\rho'(\bar{x}, y, y')]\,d\bar{x} \\ &+ \int_{x_B}^{x_B+\delta_B} f[\bar{x}, y(\bar{x}), y'(\bar{x})]\,d\bar{x} - \int_{x_A}^{x_A+\delta_A} f[\bar{x}, y(\bar{x}), y'(\bar{x})]\,d\bar{x} + \mathcal{O}(\epsilon^2)\end{aligned}$$

where $\mathcal{O}(\epsilon^2)$ refers collectively to all terms of order ϵ^2 and higher. In this expression, we have dropped terms in ϵ in the integrands of the last two integrals since these integrals are already proportional to ϵ (because the length of the integration range is proportional to ϵ). Thus terms in ϵ in the integrand will contribute only terms of order ϵ^2. We now expand the first integral through order ϵ and evaluate the last two integrals. We find that

$$\begin{aligned}\bar{I}(\epsilon) = &\int_{x_A}^{x_B} f[\bar{x}, y(\bar{x}), y'(\bar{x})]\,d\bar{x} \\ &+ \epsilon\int_{x_A}^{x_B}\left[\rho(\bar{x}, y, y')\frac{\partial}{\partial y}f(\bar{x}, y, y') + \rho'(\bar{x}, y, y')\frac{\partial}{\partial y'}f(\bar{x}, y, y')\right]d\bar{x} \\ &+ \delta_B[f(\bar{x}, y, y')]_{\bar{x}=x_B} - \delta_B[f(\bar{x}, y, y')]_{\bar{x}=x_A} + \mathcal{O}(\epsilon^2)\,.\end{aligned}$$

Integrating the last term in the second integral by parts and using Eqs. (2.33), (2.35a), and (2.35b), we get

$$\begin{aligned}\bar{I}(\epsilon) = &\int_{x_A}^{x_B} f(\bar{x}, y, y')\,d\bar{x} \\ &+ \epsilon\int_{x_A}^{x_B}\left[\frac{\partial f}{\partial y} - \frac{d}{d\bar{x}}\frac{\partial f}{\partial y'}\right]\left[\eta(\bar{x}, y, y') - \frac{dy}{d\bar{x}}\xi(\bar{x}, y, y')\right]d\bar{x} \\ &+ \epsilon\left\{\frac{\partial f}{\partial y'}\eta(\bar{x}, y, y') + \left[f(\bar{x}, y, y') - y'\frac{\partial f}{\partial y'}\right]\xi(\bar{x}, y, y')\right\}_{x_A}^{x_B} + \mathcal{O}(\epsilon^2)\,.\end{aligned}$$

But the first integral is just I (remember that the variable of integration can be called either x or $\bar{x}$ since it is a dummy variable), so we may write

$$\begin{aligned}\bar{I}(\epsilon) - I = &\,\epsilon\int_{x_A}^{x_B}\left[\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'}\right](\eta - y'\xi)\,dx \\ &+ \epsilon\left[\frac{\partial f}{\partial y'}\eta + \left(f - y'\frac{\partial f}{\partial y'}\right)\xi\right]_{x_A}^{x_B} + \mathcal{O}(\epsilon^2)\,,\end{aligned} \tag{2.36}$$

where we have written the dummy independent variable as x throughout.

Note that if in Eq. (2.36) we set $\xi \equiv 0$ (that is, $x = \bar{x}$) and require η to vanish at both x_A and x_B, then we find

$$\bar{I}(\epsilon) - \bar{I}(\epsilon = 0) = \epsilon \int_{x_A}^{x_B} \left[\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right] \eta \, dx + \mathcal{O}(\epsilon^2) , \tag{2.37}$$

since $\bar{I}(\epsilon = 0) = I$. If we require that

$$\frac{\bar{I}(\epsilon) - \bar{I}(\epsilon = 0)}{\epsilon} \to 0$$

as $\epsilon \to 0$, then the integral on the right-hand side of Eq. (2.37) must vanish. From this we conclude that

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0 .$$

This is just the basic Euler-Lagrange equation [Eq. (2.3)] derived in Section 2.2. Thus we see that if we require that the rate of change of $\bar{I}$ with respect to ϵ vanish for variations of the type $\xi \equiv 0$, $\eta|_{x_A} = 0 = \eta|_{x_B}$, then we obtain the same differential equation for y as we did in Section 2.2. The reader should be able to see the reason for this result.

The generalization of Eq. (2.36) to the case of several dependent variables, $y_1, y_2, \cdots, y_n$, is straightforward, just as it was in Section 2.2. The reader can show that in this case, if

$$I = \int_{x_A}^{x_B} f(x, y_1, y_2, \cdots, y_n; y_1', y_2', \cdots, y_n') \, dx , \tag{2.38}$$

we have

$$\begin{aligned} \bar{I}(\epsilon) - \bar{I}(\epsilon = 0) = \epsilon \int_{x_A}^{x_B} \sum_{i=1}^{n} \left[\frac{\partial f}{\partial y_i} - \frac{d}{dx} \frac{\partial f}{\partial y_i'} \right] (\eta_i - y_i' \xi) \, dx \\ + \epsilon \left[\sum_{i=1}^{n} \frac{\partial f}{\partial y_i} \eta_i + \left(f - \sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'} \right) \xi \right]_{x_A}^{x_B} + \mathcal{O}(\epsilon^2) , \end{aligned} \tag{2.39}$$

where we have written

$$\bar{y}_i(x) = y_i(x) + \epsilon \eta_i(x, y_1, \cdots, y_n; y_1', \cdots, y_n') ,$$

$$\bar{x} = x + \epsilon \xi(x, y_1, \cdots, y_n; y_1', \cdots, y_n') .$$

Again, if we assume $\xi \equiv 0$, $\eta_i|_{x_A} = 0 = \eta_i|_{x_B}$ for all i, then we obtain the Euler-Lagrange equations for n dependent variables [Eq. (2.5)],

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial y_i'}\right) = 0 \qquad i = 1, 2, \cdots, n .$$

A particularly interesting situation arises when the functions $y_i(x)$ which we are varying are solutions to the Euler-Lagrange equations. Then, according

to Eq. (2.5), the integral in Eq. (2.39) vanishes, and we have simply

$$\bar{I}(\epsilon) - \bar{I}(\epsilon = 0) = \epsilon\left[\sum_{i=1}^{n} \frac{\partial f}{\partial y_i'}\eta_i + \left(f - \sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'}\right)\xi\right]_{x_A}^{x_B}. \tag{2.40}$$

Now we have already discovered that at least in one special case the Euler-Lagrange equations can be manipulated in such a way as to produce a constant of the motion or first integral. This was the case when $f(x, y, y')$ did not depend *explicitly* on x, but only implicitly, through y and y'. In that case we found by direct calculation that

$$y' \frac{\partial f}{\partial y'} - f = \text{const}, \tag{2.41a}$$

and, generally, in the case of several dependent variables,

$$\sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'} - f = \text{const}. \tag{2.41b}$$

If f is equal to the Lagrangian function L and if the independent variable is the time t, then when we denote differentiation with respect to time by a dot, this last equation becomes

$$\sum_{i=1}^{n} \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L = \text{const}, \tag{2.42}$$

where the q_i are the generalized coordinates. As was pointed out in Section 2.5, in most cases of physical interest the constant in Eq. (2.42) is the conserved total energy of the system.

The essential element in the derivation of Eq. (2.41a) [and also Eq. (2.41b)] was the fact that f did not depend explicitly on x. Another way of obtaining Eq. (2.41a) is to show that the integral I is *invariant* under the transformation

$$\bar{x} = x + \epsilon\xi\,, \qquad \bar{y} = y(x) + \epsilon\eta \tag{2.43}$$

with

$$\xi = 1\,, \qquad \eta = 0\,. \tag{2.44}$$

By invariance, we mean simply that

$$\bar{I}(\epsilon) = \bar{I}(\epsilon = 0) = I\,. \tag{2.45}$$

Eq. (2.45) can be derived as follows. If $f(x, y, y') = F(y, y')$, then

$$I = \int_{x_A}^{x_B} F(y, y')\, dx$$

and

$$\bar{I}(\epsilon) = \int_{\bar{x}_A}^{\bar{x}_B} F[\bar{y}(\bar{x}), \bar{y}'(\bar{x})]\, d\bar{x}\,.$$

Following the prescription for $\bar{x}$ and $\bar{y}$, we have

$$\bar{x}_A = x_A + \epsilon\,, \qquad \bar{x}_B = x_B + \epsilon\,,$$

and

$$\bar{y}(\bar{x}) = y(\bar{x} - \epsilon)\,.$$

Thus

$$\bar{I}(\epsilon) = \int_{x_A+\epsilon}^{x_B+\epsilon} F\left[y(\bar{x} - \epsilon), \frac{d}{d\bar{x}} y(\bar{x} - \epsilon)\right] d\bar{x}\,.$$

But

$$\frac{d}{d\bar{x}} y(\bar{x} - \epsilon) = \frac{dy(\bar{x} - \epsilon)}{d(\bar{x} - \epsilon)}\,,$$

so that

$$\bar{I}(\epsilon) = \int_{x_A+\epsilon}^{x_B+\epsilon} F\left[y(\bar{x} - \epsilon), \frac{dy(\bar{x} - \epsilon)}{d(\bar{x} - \epsilon)}\right] d\bar{x}\,.$$

If we make the change of variable $x = \bar{x} - \epsilon$, then

$$\bar{I}(\epsilon) = \int_{x_A}^{x_B} F\left[y(x), \frac{dy(x)}{dx}\right] dx = \bar{I}(\epsilon = 0) = I\,.$$

Thus I is *invariant* under the set of transformations defined by Eqs. (2.43) and (2.44). But this means that the difference $\bar{I}(\epsilon) - I$ must vanish, so according to Eq. (2.40), we obtain

$$\epsilon\left[\frac{\partial f}{\partial y'}\eta + \left(f - y'\frac{\partial f}{\partial y'}\right)\xi\right]_{x_A}^{x_B} = 0\,.$$

Using Eq. (2.44), we can reduce this to

$$\left(f - y'\frac{\partial f}{\partial y'}\right)_{x=x_A} = \left(f - y'\frac{\partial f}{\partial y'}\right)_{x=x_B}\,.$$

Since x_A and x_B are arbitrary, we must have

$$f - y'\frac{\partial f}{\partial y'} = \text{const},$$

which is exactly the same first integral derived in Section 2.2 by a very different method. Thus we see that the existence of a first integral (or constant of the motion if we have in mind a physical problem) is related to the invariance of I under a transformation of the type discussed in this section. This important result is a special case of a powerful theorem due to Emmy Noether.

Theorem 2.1. If

$$I = \int_{x_A}^{x_B} f(x, y_1, \cdots, y_n;\, y_1', \cdots, y_n')\, dx$$

is invariant under the infinitesimal transformation generated by

$$\bar{x} = x + \epsilon\xi(x, y_1, \cdots, y_n; y_1', \cdots, y_n') ,$$

$$\bar{y}_i(x) = y_i(x) + \epsilon\eta_i(x, y_1, \cdots, y_n; y_1', \cdots, y_n') ,$$

in the sense that

$$I(\epsilon) = \int_{x_A}^{x_B} f[\bar{x}, \bar{y}_1(\bar{x}), \cdots, \bar{y}_n(\bar{x}); \bar{y}_1'(\bar{x}), \cdots, \bar{y}_n'(\bar{x})]\, d\bar{x} = I ,$$

then there exists a first integral of the related Euler-Lagrange equations which is given by

$$\sum_{i=1}^{n} \frac{\partial f}{\partial y_i'}\eta_i + \left(f - \sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'}\right)\xi = \text{const.} \tag{2.46}$$

Proof. Since $y_1, \cdots, y_n$ satisfy the Euler-Lagrange equations, by Eq. (2.40) we have

$$\bar{I}(\epsilon) - I = \epsilon\left[\sum_{i=1}^{n} \frac{\partial f}{\partial y_i'}\eta_i + \left(f - \sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'}\right)\xi\right]_{x_A}^{x_B} + \mathcal{O}(\epsilon^2);$$

since I is assumed to be invariant, we have $\bar{I}(\epsilon) - I = 0$, so that

$$\left[\sum_{i=1}^{n} \frac{\partial f}{\partial y_i'}\eta_i + \left(f - \sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'}\right)\xi\right]_{x=x_A} = \left[\sum_{i=1}^{n} \frac{\partial f}{\partial y_i'}\eta_i + \left(f - \sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'}\right)\xi\right]_{x=x_B} .$$

Since x_A and x_B are arbitrary, we conclude that

$$\sum_{i=1}^{n} \frac{\partial f}{\partial y_i'}\eta_i + \left(f - \sum_{i=1}^{n} y_i' \frac{\partial f}{\partial y_i'}\right)\xi = \text{const},$$

which is the required result.

The first integral obtained in the case that f does not depend explicitly on x is an example of this result. Note that the use of the term first integral is justified since Eq. (2.46) contains only *first* derivatives of the y_i, whereas the Euler-Lagrange equations are *second-order* differential equations. We have already mentioned that if f is the Lagrangian of a dynamical system,

$$f = L(t, q_1, \cdots, q_n; \dot{q}_1, \cdots, \dot{q}_n) \equiv T - V ,$$

then this first integral corresponds to the conservation of energy in the case that the Lagrangian does not depend *explicitly* on time.

Another interesting example which is furnished by a classical mechanical Lagrangian is the case where V, the potential energy, depends only on the vectors joining pairs of particles. For example, in a two-particle system, this would mean that

$$L = \frac{m_1}{2}(\dot{x}_1^2 + \dot{y}_1^2 + \dot{z}_1^2) + \frac{m_2}{2}(\dot{x}_2^2 + \dot{y}_2^2 + \dot{z}_2^2) - V(\mathbf{r}_1 - \mathbf{r}_2) .$$

Such a Lagrangian is invariant under the transformation

$$\bar{t} = t + \epsilon\tau\,, \qquad \bar{x}_1 = x_1 + \epsilon\xi_1\,, \qquad \bar{x}_2 = x_2 + \epsilon\xi_2\,, \qquad \bar{y}_1 = y_1 + \epsilon\eta_1\,,$$
$$\bar{y}_2 = y_2 + \epsilon\eta_2\,, \qquad \bar{z}_1 = z_1 + \epsilon\zeta_1\,, \qquad \bar{z}_2 = z_2 + \epsilon\zeta_2\,,$$

where

$$\tau = 0\,, \qquad \xi_1 = 1\,, \qquad \xi_2 = 1\,,$$
$$\eta_1 = \eta_2 = \zeta_1 = \zeta_2 = 0\,.$$

Using Eq. (2.46), we have immediately

$$\sum_{i=1}^{2} \frac{\partial L}{\partial \dot{x}_i}\xi_i = \text{const}, \qquad \text{or} \qquad \frac{\partial L}{\partial \dot{x}_1} + \frac{\partial L}{\partial \dot{x}_2} = m_1\dot{x}_1 + m_2\dot{x}_2 = \text{const}.$$

This simply states that the x-component of the total linear momentum of the system is a constant of the motion. Similarly, invariance under translation of the y-coordinates leads to the same result for the y-component of the total linear momentum, and invariance under translation of the z-coordinates leads to conservation of the z-component of the total linear momentum. Thus we see that conservation of *linear* momentum is an immediate consequence of the *translational* invariance of the system, defined in terms of the integral I, appearing in Hamilton's principle.

As a final example, let us consider the Lagrangian of a particle moving in a central potential $V(r)$. We have for L,

$$L = \frac{m}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - V(r)\,, \qquad r = (x^2 + y^2 + z^2)^{1/2}\,. \tag{2.47}$$

Consider now an arbitrary rotation about the z-axis. This corresponds to the transformation

$$\bar{t} = t\,, \qquad \bar{x} = x\cos\epsilon + y\sin\epsilon\,, \qquad \bar{y} = -x\sin\epsilon + y\cos\epsilon\,, \qquad \bar{z} = z\,, \tag{2.48}$$

as may be seen by looking back at Eq. (1.15). The corresponding values of τ, ξ, η and ζ are

$$\tau = 0\,, \qquad \xi = y\,, \qquad \eta = -x\,, \qquad \zeta = 0\,,$$

since the infinitesimal transformation corresponding to $\bar{x}$ and $\bar{y}$ is [letting ϵ become small in Eq. (2.48)]

$$\bar{x} = x + \epsilon y\,, \qquad \bar{y} = y - \epsilon x.$$

Substituting Eqs. (2.48) into the Lagrangian of Eq. (2.47), we see that it is invariant under rotations about the z-axis. Equation (2.46) then gives

$$\frac{\partial L}{\partial \dot{x}}\xi + \frac{\partial L}{\partial \dot{y}}\eta = \text{const}\,, \qquad \text{or} \qquad x\frac{\partial L}{\partial \dot{y}} - y\frac{\partial L}{\partial \dot{x}} = \text{const}\,.$$

But

$$\frac{\partial L}{\partial \dot{x}} = p_x , \qquad \frac{\partial L}{\partial \dot{y}} = p_y ,$$

so we get

$$xp_y - yp_x = (\mathbf{r} \times \mathbf{p})_z = L_z = \text{const},$$

which is a statement of the conservation of angular momentum about the z-axis. In an identical manner, it follows from the invariance of the system under rotations about the x and y axis that L_x and L_y are also conserved in the motion of the system; i.e., the *total* angular momentum of the system is conserved.

Thus, from the examples considered, we see that one of the great advantages of being able to formulate classical mechanics in terms of Hamilton's principle is that from this viewpoint we are easily able to relate conservation laws to the existence of underlying symmetries under which the quantity

$$I = \int_{t_A}^{t_B} L\, dt$$

is invariant.

PROBLEMS

1. *The shortest line (polar coordinates).* Find the shortest distance between two points using polar coordinates, i. e., using them as a line element:
$$ds^2 = dr^2 + r^2 d\theta^2 .$$
2. *The shortest line (three dimensions).* Prove that the shortest distance between two points in three-dimensional space is a straight line. Take the origin and (a, b, c) as the two points and actually determine the equation of the line.
3. *Geodesics on a sphere.* On a sphere of fixed radius a, the line element is given by
$$ds^2 = a^2 d\theta^2 + a^2 \sin^2 \theta d\varphi^2 .$$
Determine the equation of the curve that extremizes the distance between two points. Is the extremum a minimum, a maximum, or what? [*Hint*: Let θ be the independent variable so the integrand of the integral to be extremized is independent of the *dependent* variable φ.] Now prove: $\varphi = \alpha - \sin^{-1}(k \cot \theta)$, where α and k are constants. Then change from the spherical coordinates to Cartesian coordinates to see that this is the equation of a plane through the origin; its intersection with the sphere $r = a$ is the geodesic—a great circle—on the surface of the sphere.
4. *Inverse isoperimetric problem* ("isoareametric" problem). Prove that of all simple closed curves enclosing a given area, the least perimeter is possessed by the circle.
5. Demonstrate the following reciprocity relationship for the simple isoperimetric problem: The particular function which renders
$$I = \int_{x_1}^{x_2} f(x, y, y')\, dx$$

an extremum with respect to functions which give

$$J = \int_{x_1}^{x_2} g(x, y, y')\, dx$$

a prescribed value, also renders J an extremum with respect to functions which give I a prescribed value if the Lagrange multiplier $\lambda \neq 0$.

6. In all our discussions so far on finding the function f for which $I = \int_{x_1}^{x_2} f\,dx$ is an extremum, it has been assumed that f depends only on x, y, and y':

$$f = f(x, y, y') .$$

Show that if f also involves the second derivative of y with respect to x, $f = f(x, y, y', y'')$, then for fixed endpoints and prescribed y' at the endpoints, the Euler-Lagrange equation is

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) + \frac{d^2}{dx^2}\left(\frac{\partial f}{\partial y''}\right) = 0 .$$

What continuity assumptions must be made about derivatives of y?

7. Fermat's principle states: *If the velocity of light is given by the continuous function $u = u(y)$, the actual light path connecting the points (x_1, y_1) and (x_2, y_2) in a plane is one which extremizes the time integral*

$$I = \int_{(x_1, y_1)}^{(x_2, y_2)} \frac{ds}{u} .$$

(Actually, refinements are needed to make this formulation of Fermat's principle hold for all cases.)

a) Derive Snell's law from Fermat's principle; that is, prove that $\sin \phi / u = \text{const}$, where ϕ is the angle shown in Fig. 2.6.

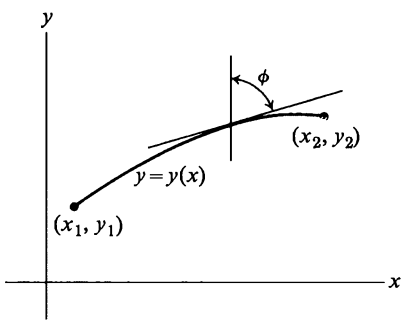

Fig. 2.6

b) Suppose that light travels in the xy-plane in such a way that its speed is proportional to y; then prove that the light rays emitted from any point are circles with their centers on the x-axis.

8. *Brachistochrone problem for a central gravitational force.* A particle of mass m is to move under the gravitational attraction of another stationary spherical mass M from a point at a finite distance $r = R_0$ from the center of M to a point on the surface of M. We want to determine the path such that the particle moving along

it will travel between the two fixed endpoints in a time less than the time of transit along any other path. Taking for the two endpoints ($\theta = 0$, $r = R_0$) and ($\theta = \alpha$,

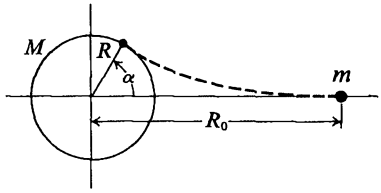

Fig. 2.7

$r = R$) as shown in Fig. 2.7, set up in polar coordinates the equation that determines the path and solve for θ as a function of r. The best one can do analytically is to express θ as an elliptic integral with a variable upper limit and lower limit chosen to satisfy one of the boundary conditions. The determination of this path tells us how best to design interplanetary mail chutes.

9. Show that if ψ and ψ^* are taken as two independent field variables (dependent variables) the real Lagrangian density

$$\mathscr{L} = \frac{\hbar^2}{2m}\nabla\psi\cdot\nabla\psi^* + \mathrm{V}\psi\psi^* - \frac{i\hbar}{2}\left(\psi^*\dot{\psi} - \psi\dot{\psi}^*\right),$$

where $\dot{\psi} = \partial\psi/\partial t$ leads to the time-dependent Schrödinger equation

$$H\psi = \left(\frac{-\hbar^2}{2m}\nabla^2 + \mathrm{V}\right)\psi = i\hbar\frac{\partial\psi}{\partial t}$$

and its complex conjugate.

10. a) A charged scalar meson is described by two real fields $\varphi_1(x_\mu)$, $\varphi_2(x_\mu)$, where $x_\mu = (x, y, z, it)$, (natural units: $c = \hbar = 1$). The Lagrange density for such a meson in an external electromagnetic field, described by the vector potential $A_\mu(x)$, is

$$\mathscr{L} = -\frac{1}{2}\left(\frac{\partial\varphi_1}{\partial x_\mu}\frac{\partial\varphi_1}{\partial x_\mu} + \frac{\partial\varphi_2}{\partial x_\mu}\frac{\partial\varphi_2}{\partial x_\mu}\right) - \frac{m^2}{2}\left(\varphi_1^2 + \varphi_2^2\right)$$
$$+ eA_\mu\left[\varphi_2\frac{\partial\varphi_1}{\partial x_\mu} - \varphi_1\frac{\partial\varphi_2}{\partial x_\mu}\right] + \frac{1}{2}e^2A_\mu A_\mu(\varphi_1^2 + \varphi_2^2)\,.$$

Here we use the Einstein summation convention (summation over repeated indices). Find the Euler-Lagrange equations for φ_1 and φ_2 corresponding to this Lagrangian; e and m are constants; A_μ is a given function of x.

b) Sometimes the complex fields φ and φ^* are introduced in place of the two real fields φ_1 and φ_2:

$$\varphi(x_\mu) = \frac{1}{\sqrt{2}}\left[\varphi_1(x_\mu) - i\varphi_2(x_\mu)\right],$$
$$\varphi^*(x_\mu) = \frac{1}{\sqrt{2}}\left[\varphi_1(x_\mu) + i\varphi_2(x_\mu)\right].$$

Show that the Lagrangian density in terms of these two new complex dependent field variables in free space ($A_\mu = 0$) is

$$\mathscr{L}_0 = -\left(\frac{\partial\varphi^*}{\partial x_\mu}\frac{\partial\varphi}{\partial x_\mu} + m^2\varphi^*\varphi\right).$$

Now determine the Euler-Lagrange equations for this Lagrangian density (either independently or from part (a) of the problem by transformation of φ_1 and φ_2 into φ and φ^*). You will find that the Euler-Lagrange equations are

$$(\Box^2 - m^2)\varphi = 0\,, \qquad \text{and} \qquad (\Box^2 - m^2)\varphi^* = 0\,,$$

the Klein-Gordon equation which describes a spinless particle in free space. Here the two equations describe the two charged π-mesons.

11. The Lagrangian density $\mathscr{L}$, which generates a given set of Euler-Lagrange equations, is not unique. Prove this result by showing that adding a divergence to $\mathscr{L}$ does not alter the Euler-Lagrange equations. That is, let

$$\mathscr{L}' = \mathscr{L} + \sum_k \frac{\partial f_k}{\partial x_k}\,,$$

where

$$\mathscr{L} = \mathscr{L}\left(x_k,\ \varphi_j,\ \frac{\partial \varphi_j}{\partial x_k}\right);$$
$$f_k = f_k(\varphi_j)\ ;$$

$j = 1, \cdots, m$ indexes the dependent field variables, and
$k = 1, \cdots, n$ indexes the independent variables.

Now prove that

$$\frac{\delta \mathscr{L}'}{\delta \varphi_j} = \frac{\delta \mathscr{L}}{\delta \varphi_j}\,,$$

where we define

$$\frac{\delta \mathscr{L}}{\delta \varphi_j} \equiv \frac{\partial \mathscr{L}}{\partial \varphi_j} - \sum_i \frac{\partial}{\partial x_i} \frac{\partial \mathscr{L}}{\partial(\partial \varphi_j/\partial x_i)}\,.$$

12. A Lagrangian density for classical electrodynamics is

$$\mathscr{L} = -\frac{1}{16\pi} \sum_{i,j} F_{ij}F_{ij} + \frac{1}{c} \sum_i J_i A_i$$

with

$$F_{ij} = \frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j}\,.$$

The four-vectors $\{A_i\}$ and $\{J_i\}$ are defined by

$$\{A_i\} \equiv \{A_1, A_2, A_3, i\varphi\} \equiv \{\mathbf{A}, i\varphi\}\,,$$
$$\{J_i\} \equiv \{j_1, j_2, j_3, ic\rho\} \equiv \{\mathbf{j}, ic\rho\}\,,$$

where $\mathbf{A}$ is the vector potential ($\mathbf{H} = \nabla \times \mathbf{A}$), φ is the scalar potential

$$\left(\mathbf{E} = -\nabla\varphi - \frac{1}{c}\frac{\partial \mathbf{A}}{dt}\right),$$

$\mathbf{j}$ is the current vector, and ρ is the charge density (see Problem 1–13). Show that this Lagrangian density can be written in a more conventional form as

$$\mathscr{L} = \frac{1}{8\pi}(\mathbf{E}^2 - \mathbf{H}^2) + \frac{1}{c}\mathbf{j}\cdot\mathbf{A} - \rho\varphi\,.$$

Using Lagrange's equations with **A** and φ as dependent variables, show that this Lagrangian density leads to the familiar Maxwell equations.

13. Prove that a particle moving under gravity in a plane from a fixed point P to a vertical line L will reach the line in minimum time by following the cycloid from P to L that intersects L at right angles.

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CHAPTER 3

VECTORS AND MATRICES

INTRODUCTION

The theory of vector spaces is to quantum mechanics what calculus is to classical mechanics. In this chapter we shall deal principally with finite-dimensional vector space, but in such a way that the generalization to infinite-dimensional spaces (Chapters 5 and 8), which forms the mathematical language of quantum theory, is natural and entails little revision. In fact, almost all the theorems proved in this chapter for finite-dimensional spaces will hold in the infinite-dimensional generalization.

Our discussion of vector spaces will be a good deal more abstract and formal than the discussion of vectors in Chapter 1. We now concentrate primarily on the algebraic aspects of vectors. However, much of the geometric language is retained and given a precise algebraic interpretation.

3.1 GROUPS, FIELDS, AND VECTOR SPACES

The definition of an abstract vector to be given in this chapter depends very much on the properties of scalars, even though the vectors will turn out to obey quite different combinatorial rules than the scalars. For example, the components of a vector are scalars. And a scalar may operate on a vector: e.g., a scalar may multiply a vector and thus change its magnitude.

It is customary to speak of the vectors, or the vector space, as being defined "over a field of scalars." The word *field* is just shorthand for the list of axioms that the *scalars* obey. We usually take these rules for granted, but we shall state them explicitly here. In the process, it will be convenient first to introduce another algebraic structure called a *group*. Like a field, a group is a system consisting of a set of elements and rules for combining the members of this set.

Suppose, then, that G is a set of elements, and let $\cdot$ be an operation defined on the elements of G such that, if a and $b \in G$, then $(a \cdot b) \in G$ as well.* Such an operation is said to be *closed*. Then we define a group as follows.

Definition 3.1. A group is a system $\{G, \cdot\}$, consisting of the set G and a single *closed* operation $\cdot$, which satisfies the following three axioms:

1. If $a, b, c \in G$, then $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity).
2. There exists an identity element $e \in G$ such that for all $a \in G$, $a \cdot e = e \cdot a = a$.

* The symbol $\in$ means "belongs to" or "is a member of."

3. For every $a \in G$, there exists an inverse element in G, denoted a^{-1}, such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.*

A group is said to be *abelian* if it is commutative, that is, if $a \cdot b = b \cdot a$, where $a, b \in G$. Often the dot $(\cdot)$ is omitted if it is interpreted as the usual multiplication, but for some sets the dot will be interpreted as $+$ (addition). As used in our definition, it denotes any operation that combines two elements of the set to give a third element of the set, just so the rule of combination satisfies the three axioms.

The theory of groups is very important in physics. For instance, in the study of quantum mechanics one often hears, "It can be shown by group theory that . . ." We shall discuss this subject in its own right in Chapter 10.

Examples of Groups.

1. The set of all real numbers excluding 0 forms a group with respect to the usual multiplication $(\cdot)$. The identity is 1, and for each x the inverse is $x^{-1} \equiv 1/x$. This group is abelian.
2. The integers with addition as operation compose another abelian group; 0 is the identity element, and the inverse is $n^{-1} \equiv -n$.
3. The set of all two-dimensional rotation matrices [see Eq. (1.14)]

$$r(\phi) \equiv \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix}, \tag{3.1}$$

is an abelian, *continuous* group—continuous because there is an infinite number of elements defined by the continuous variation of the parameter φ. Multiplication of group elements (matrices) was defined briefly in Eq. (1.22). Matrix multiplication will be discussed in detail in this Chapter. The unit element of this group is $r(0)$ and the inverse of $r(\phi)$ is $r(-\phi)$.

4. The set of three-dimensional rotations is a noncommutative continuous group. The reader may demonstrate this by noting that the net result of two consecutive rotations in three dimensions is dependent on their order (see Figs. 3.1 and 3.2).
5. The set of permutations of n objects is a group. For example, the $3! = 6$ possible permutations of 3 elements may be written as

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} = I, \quad \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} = F, \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = D$$
$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} = A, \quad \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} = C, \quad \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} = B. \tag{3.2}$$

For example, the permutation F means that 1 is replaced by 3, 2 is replaced by 1, and 3 is replaced by 2. The product of two permutations, denoted by $*$, is the single permutation that accomplishes what two applied successively, from right to left would accomplish, e.g.,

$$A * C = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = D.$$

* This is not a minimal definition from a mathematical point of view. Such a definition will be given in Chapter 10.

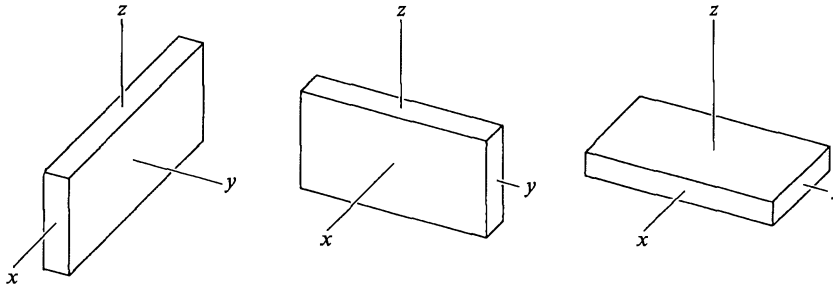

Fig. 3.1 A rotation about the z-axis followed by one about the y-axis.

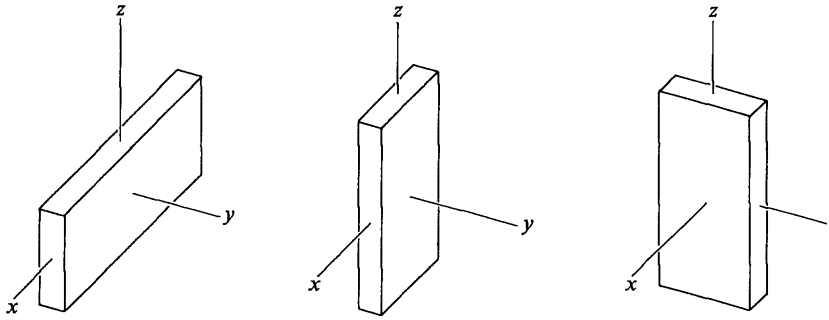

Fig. 3.2 A rotation about the y-axis followed by one about the z-axis.

This is obtained as follows: First the permutation C taken $1 \to 1$, $2 \to 3$, $3 \to 2$. Then A taken $1 \to 2$, $2 \to 1$, $3 \to 3$. The combined effect of first C and then A may be written:

$$1 \to 1 \to 2\,, \qquad 2 \to 3 \to 3\,, \qquad 3 \to 2 \to 1\,.$$

The net result of the two permutations acting successively, from right to left, is thus: $1 \to 2$, $2 \to 3$, $3 \to 1$; that is, the single permutation D. Similarly,

$$C * A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} = F\,. \tag{3.3}$$

Thus this group is not abelian. The 36 possible multiplications for the permutation group of three objects are shown in Table 3.1.

Table 3.1

	I	A	B	C	D	F
I	I	A	B	C	D	F
A	A	I	F	D	C	B
B	B	D	I	F	A	C
C	C	F	D	I	B	A
D	D	B	C	A	F	I
F	F	C	A	B	I	D

(3.4)

Now we define a field in terms of a group.

Definition 3.2. A *field* is a system $\{F, +, \cdot\}$ satisfying the three axioms:

1. The system $\{F, +\}$ is an abelian group with the identity element 0.
2. Let F' be the set of all $x \in F$ except $x = 0$. Then system $\{F', \cdot\}$ is an abelian group with the identity element e.
3. Let $a, b, c \in F$; then $a \cdot (b + c) = a \cdot b + a \cdot c$ (distributivity of $\cdot$ with respect to $+$).

Examples of a Field. The set of all rational numbers, the set of all real numbers and the set of all complex numbers with the usual multiplication and addition are the most familiar examples of a field.

Now we can define a vector space V. We assume we are given a field F; the scalars to be used are all elements of F.

Definition 3.3. A *vector space* over a field F is a set of elements, $V(F)$, called vectors, which satisfy the following axioms:

1. There exists an operation $(+)$ on the vectors such that $\{V(F), +\}$ forms an abelian group; the identity element is denoted by 0.
2. For every $\alpha \in F$ and $x \in V(F)$, there exists a vector $\alpha x \in V(F)$; furthermore,

a) $\alpha(\beta x) = (\alpha\beta)x$, b) $1(x) = x$, for all x.

c) $\alpha(x + y) = \alpha x + \alpha y$, d) $(\alpha + \beta)x = \alpha x + \beta x$.

A vector space $V(F)$ is called a real or a complex vector space according to whether F is the field of real numbers or of complex numbers.

We shall adopt the following notational conventions:

a) Vector spaces will be written $V_n(F)$, where the subscript denotes the dimension of the space (to be defined shortly) and the quantity in parentheses is the field over which the vector space is defined. Where the dimensionality and/or the field are apparent from the context, we may write just V_n or V.

b) Vectors in the space will be denoted by letters at the end of the alphabet, e.g., x, y, z; they will *not* be written in bold-face type as in Chapter 1. The reader will be able to distinguish the scalar and the vector zero from the context.

c) The scalar components of vectors will be written with subscripted Greek letters, thus: $(\alpha_1, \alpha_2, \cdots, \alpha_n)$. The elements of matrices will be written with doubly subscripted Roman letters: a_{12}, etc.

d) Scalars which operate on vectors will be denoted by unsubscripted Roman or Greek letters: a, b, c, or α, β, γ. These may be either real or complex.

e) Sets of basis vectors will be denoted $\{x_i\}$, or $\{x_1, x_2, \cdots, x_n\}$. In accordance with convention b), such expressions are *not* to be interpreted as the scalar components of a single vector. For example, in this notation we write,

$$x = \sum_i \alpha_i x_i ,$$

where x is a vector whose ith component is α_i in the basis $\{x_i\}$.

Examples of Vector Spaces.

1. For a fixed positive integer n, the set of all n-tuples of complex numbers $(x = (\xi_1, \xi_2, \cdots, \xi_n))$ forms a vector space if addition of vectors and multiplication of a vector by a scalar are defined by

$$\begin{aligned} x + y &= (\xi_1, \xi_2, \cdots, \xi_n) + (\eta_1, \eta_2, \cdots, \eta_n) \\ &= (\xi_1 + \eta_1, \xi_2 + \eta_2, \cdots, \xi_n + \eta_n); \\ \alpha x &= \alpha(\xi_1, \cdots, \xi_n) = (\alpha\xi_1, \cdots, \alpha\xi_n); \end{aligned}$$

furthermore,

$$0 = (0, 0, \cdots, 0);$$

and

$$-x = (-\xi_1, -\xi_2, \cdots, -\xi_n).$$

It is easy to show that all the axioms for a vector space are satisfied. (In fact, this vector space is the prototype for all finite-dimensional vector spaces, as we shall see later.)

2. The set of all complex numbers, where we interpret $x + y$ and αx as ordinary complex numerical addition and multiplication, is a complex vector space.

3. The set P of all polynomials in a real variable t with complex coefficients is a complex vector space, provided that we interpret vector addition and scalar multiplication as ordinary addition of two polynomials and the multiplication of a polynomial by a complex number. The zero vector, 0, in P is the polynomial identically zero; P is not a finite-dimensional vector space.

As special cases of (2) and (3), we have the real vector spaces composed of the sets of all real numbers and of all polynomials in t with real coefficients, where the operations $+$ and $\cdot$ are those of ordinary arithmetic.

3.2 LINEAR INDEPENDENCE

We already have an idea of what it means to say that any two noncollinear vectors in the plane are linearly independent, and we also know that if you add any third vector in the plane, the resulting set of three vectors is linearly dependent. That is, we know that any one of the three can be expanded in terms of the other two: one cannot find three vectors in a two-dimensional space that are linearly independent. In these simple statements we have hinted at the content of several theorems that we shall prove in this section. The definition of linear independence given below will tie in neatly with these intuitive notions.

A word on summation notation: in general, we shall not use the Einstein summation convention in this chapter because it is often necessary to indicate the *range* of the summation. Thus we write $\sum_{i=1}^{n}$ to mean the sum from 1 to n. Sometimes, however, when the range of summation is perfectly clear from the context, we may omit the range, writing simply $\sum_i$.

Definition 3.4. A finite set $\{x_i\}$ of n vectors is *linearly independent* if and only if

$$\sum_{i=1}^{n} \alpha_i x_i = 0 \Longrightarrow \alpha_i = 0$$

for all i. If there exists a set of scalars $\{\alpha_i\}$ that are not all zero, such that

$$\sum_{i=1}^{n} \alpha_i x_i = 0 ,$$

then the set of vectors $\{x_i\}$ is *linearly dependent.*

The definition says that if the $\{x_i\}$ are linearly independent—as are, for example, the $\mathbf{e}_i$ unit vectors in Cartesian three-space—the zero vector can be obtained as a linear combination of them only in a trivial way, i.e., all $\alpha_i = 0$. But if the zero vector is a nontrivial linear combination of the x_i, so that not all $\alpha_i = 0$, then

$$\sum_{i=1}^{n} \alpha_i x_i = 0 \Longrightarrow x_j = -\frac{1}{\alpha_j} \sum_{i \neq j} \alpha_i x_i ,$$

where α_j is a nonzero α_i. (The symbol $\sum_{i \neq j}$ means that the sum is over all i except $i = j$.) In this case the vector x_j *depends* linearly on the others in the sense that x_j is a linear combination of the others. Note that any set of vectors containing the zero vector is linearly dependent.

We have already mentioned that any two nonzero, noncollinear vectors in the plane are linearly independent, while any three vectors in the plane are linearly dependent. Another illustration of Definition 3.4 may be found in the vector space P of polynomials mentioned earlier. Let

$$x_1(t) = 2 - t + t^2 , \qquad x_2(t) = t - t^2 + 2t^3 , \qquad x_3(t) = 1 + t^3 ,$$

be three vectors in P. If we set $\alpha_1 = \alpha_2 = 1$, and $\alpha_3 = -2$, then

$$\sum_{i=1}^{3} \alpha_i x_i = x_1 + x_2 - 2x_3 = 0 .$$

This means that x_1, x_2, and x_3 are linearly dependent, for we have found a set of α_i which are not all zero such that $\sum_i \alpha_i x_i$ vanishes. However, the infinite set of vectors

$$\begin{aligned} x_0(t) &= 1 \\ x_1(t) &= t \\ &\vdots \\ x_n(t) &= t^n \\ &\vdots \end{aligned}$$

is a linearly independent set because if for any n

$$\sum_{i=0}^{n} \alpha_i x_i = 0 ,$$

then we have the polynomial identity:

$$\alpha_0 + \alpha_1 t + \cdots + \alpha_n t^n = 0 ,$$

which implies

$$\alpha_i = 0 \qquad \text{for} \quad 0 \leq i \leq n ,$$

since this must hold for all t.

The most important result involving the notions of linear dependence and independence is contained in the following theorem.

Theorem 3.1. The set of nonzero vectors $\{x_i\}$ is linearly dependent if and only if some x_i, $2 \leq i \leq n$, is a linear combination of the preceding ones.

Proof. We first prove the necessity of the condition. We assume that the x_i are linearly dependent. This means that for some i (and let us pick the smallest such i and call it k) we know that

$$\sum_{i=1}^{k} \alpha_i x_i = 0 ,$$

where $k \leq n$ and not all the $\alpha_i = 0$. Now we know $\alpha_k \neq 0$, because if $\alpha_k = 0$, the set $\{x_1, \cdots, x_{k-1}\}$ would be linearly dependent, contrary to the definition of k as the *smallest* value of i for which the vectors $x_1, \cdots, x_k$ are linearly dependent. Therefore

$$x_k = -\frac{1}{\alpha_k}\left[\sum_{i=1}^{k-1} \alpha_i x_i \right] .$$

Thus x_k has been explicitly expressed as a linear combination of preceding vectors. The proof of the sufficiency of the condition is simple. If we know that

$$x_j = \sum_{i=1}^{j-1} \alpha_i x_i \qquad \text{for some } j ,$$

then letting $\alpha_j = -1$, we have

$$\sum_{i=1}^{j} \alpha_i x_i = 0 .$$

For the remaining vectors $(x_{j+1}, \cdots, x_n)$, we set $\alpha_{j+1} = \alpha_{j+2} = \cdots = \alpha_n = 0$, so that

$$\sum_{i=1}^{n} \alpha_i x_i = 0 ,$$

which implies linear dependence, since not all the α_i's are zero. QED